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## 2. Semiconductor Physics

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**Intrinsic Semiconductors - Energy band diagram - direct and indirect band gap semiconductors - Carrier concentration in intrinsic semiconductors - extrinsic semiconductors - Carrier concentration in N-type & P-type semiconductors - Variation of carrier concentration with temperature - variation of Fermi level with temperature and impurity concentration - Carrier transport in Semiconductor: random motion, drift, mobility and diffusion - Hall effect and devices - Ohmic contacts - Schottky diode.**

### Introduction

Semiconducting material has electrical conductivity between a good conductor and a good insulator. It is simply called **semiconductor**. **It is a special class of material which is very small in size and sensitive to heat, light and electricity.**

Semiconducting materials behave as insulators at low temperature and as conductors at high temperature. Moreover, these materials have two types of charge carriers i.e., **electrons and holes**.

**Germanium** and **silicon** are two important elemental semiconductors. They are used in diodes and transistors.

**Gallium arsenide (GaAs) and indium phosphide (InP)** are two important compound semiconductors. They are used in LEDs and Laser diodes.

The study of semiconducting materials is essential for engineers due to their wide applications in semiconductor devices in engineering and technology.

The invention of semiconductors opened a new branch of technology called **solid state electronics**.

It leads to the development of ICs, microprocessors, computers and supercomputers.

In short, semiconductors play a vital role in almost all advanced electronic devices.

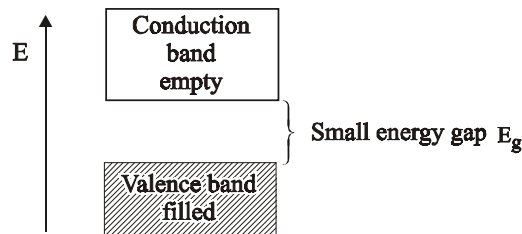
### Definition

#### Based on Electrical resistance

*Semiconductor has electrical resistance which is lesser than an insulator but more than that of a conductor. Its electrical resistivity is in the order of  $10^{-4}$  to 0.5 ohm metre.*

#### Based on Energy band

*A semiconductor has nearly an empty conduction band and almost filled valence band with a very small energy gap ( $\approx 1$  eV). (Fig 2.1)*



**Fig. 2.1** Energy band diagram of a semiconductor

### General properties of the semiconductors

- They have crystalline structure.
- Bonding between the atoms is formed by covalent bond.
- They have empty conduction band at 0K.
- They have almost filled valence band.
- The energy gap is small.
- They exhibit a negative temperature coefficient of resistance. ie., increase in temperature leads to decrease in resistance.

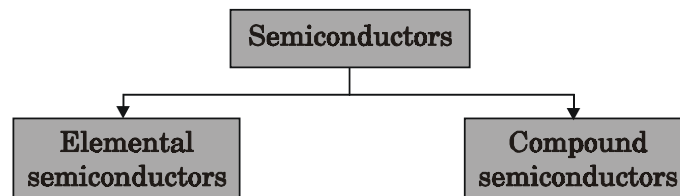
- If impurities are added to a semiconductor, its electrical conductivity increases. Further, if temperature of the semiconductor is increased, its electrical conductivity increases.

This property is in contrary to that of metals in which if temperature is increased or impurities are added, their electrical conductivity decreases.

### Classification of semiconductors

The semiconductors are classified mainly into two types based on the composition of materials. They are

- (i) Elemental semiconductors
- (ii) Compound semiconductors



### Elemental Semiconductors

(Indirect band gap semiconductor)

*The semiconductors which are made from a single element of fourth group elements in periodic table are known as elemental semiconductors.*

*They are also called as indirect band gap semiconductors.*

### Example

Two important elemental semiconductors and their energy band gaps are given in table 2.1.

**Table 2.1**

| S.No. | Element   | $E_g$ in eV |
|-------|-----------|-------------|
| 1     | Germanium | 0.72        |
| 2     | Silicon   | 1.1         |

### Compound semiconductors (Direct band gap semiconductors)

Semiconductors which are formed by combining third and fifth group or second and sixth group elements in the periodic table are known as compound semiconductors.

They are also called as **direct bandgap semiconductors**.

#### *Characteristics*

- The compound semiconductor have large forbidden gap and carrier mobility.
- They are formed by both ionic and covalent bonds.
- The recombination of electron and hole takes place directly. During this process, the light photons are emitted in visible or infrared region.

Some important compound semiconductors are given in table 2.2.

**Table 2.2**

| S.No. | Group                                                      | Compound semiconductor                                                                                 |
|-------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 1.    | Combination of third and fifth group elements (III and V)  | Gallium Phosphide (GaP)<br>Gallium Arsenide (GaAs)<br>Indium Phosphide (InP)<br>Indium Arsenide (InAs) |
| 2.    | Combination of second and sixth group elements (II and VI) | Magnesium Oxide (MgO)<br>Magnesium Silicon (MgSi)<br>Zinc Oxide (ZnO)<br>Zinc Sulphide (ZnS)           |

**Uses**

The compound semiconductors are used in photovoltaic materials, photoconductive cell, LEDs [Light Emitting Diode] and Laser diodes.

The differences between elemental and compound semiconductors are given in table 2.3.

|                  |
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| <b>Table 2.3</b> |
|------------------|

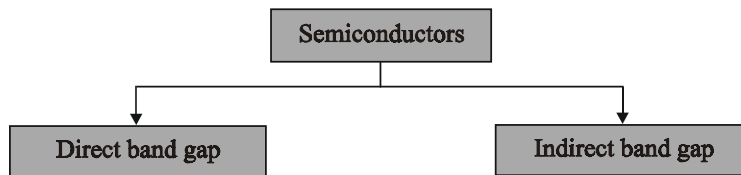
**Differences between elemental and compound semiconductors (indirect and direct band gap semiconductor)**

| S.No. | Elemental Semiconductors                                                               | Compound Semiconductors                                             |
|-------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1.    | They are made of a single element<br>Example: Germanium (Ge), Silicon (Si).            | They are made of compounds<br>Example: GaA, GaP, CdS, MgO.          |
| 2.    | They are known as <b>indirect band gap semiconductors.</b>                             | They are known as <b>direct band gap semiconductors.</b>            |
| 3.    | Electron - hole recombination takes place through traps which are present in band gap. | Electron - hole recombination takes place directly with each other. |
| 4.    | Life time of charge carriers is more due to indirect recombination.                    | Life time of charge carrier is less due to direct recombination.    |
| 5.    | Heat energy is produced during recombination.                                          | Light photons are emitted during recombination.                     |
| 6.    | They carry more current.                                                               | They carry less current.                                            |
| 7.    | They are used for making diodes and transistors.                                       | They are used for making LED's and Laser diodes.                    |

**2.1****DIRECT AND INDIRECT BAND GAP SEMICONDUCTORS**

Semiconductors are also classified into

- (a) **Direct band gap semiconductor**
- (b) **Indirect band gap semiconductor.**



The electrons and holes in a semiconductor have energy and momentum. The momentum ( $k$ ) depends on energy ( $E$ ). A plot of energy versus momentum is as shown in fig. 2.2

The lower curves represent energy and momentum values of holes in valence band of semiconductor. Similarly upper curves denote corresponding values for electrons in conduction band.

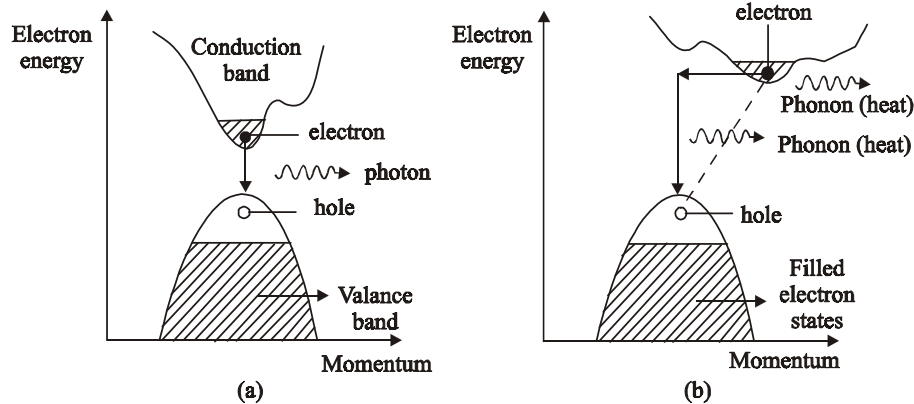
In direct band gap semiconductor, the energy maximum of valence band and the energy minimum of the conduction band are having same momentum value.

During the recombination of electron from conduction band with hole in valence band, the momentum of the electrons remains virtually constant. The energy equal to band gap energy is released as **light photon**.

But, in the case of indirect band gap semiconductor, the maximum energy of valence band and minimum energy of conduction band are having different values of momentum.

During recombination, electron firstly loses momentum such that it has momentum equal to the momentum corresponding to energy maximum of valence band.

To conserve the momentum, emission of third particle known as a phonon is generated. Thus, in this type of recombination phonon is produced.



**Fig. 2.2 Energy - momentum diagram**

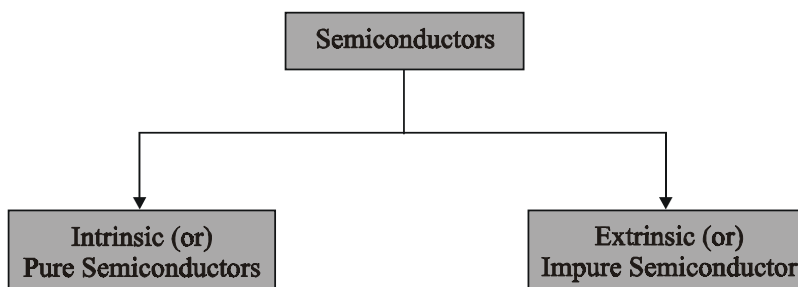
**(a) Direct band gap semiconductor**

**(b) Indirect band gap semiconductor**

## Types of Semiconductors

When a suitable impurity is added to a pure semiconductor, its electrical conductivity changes. Based on this property, the semiconductors are classified into two types. They are

- (i) ***Intrinsic semiconductor or pure semiconductor***
- (ii) ***Extrinsic semiconductor or impure semiconductor or Doped semiconductor***



## 2.2 INTRINSIC SEMICONDUCTORS

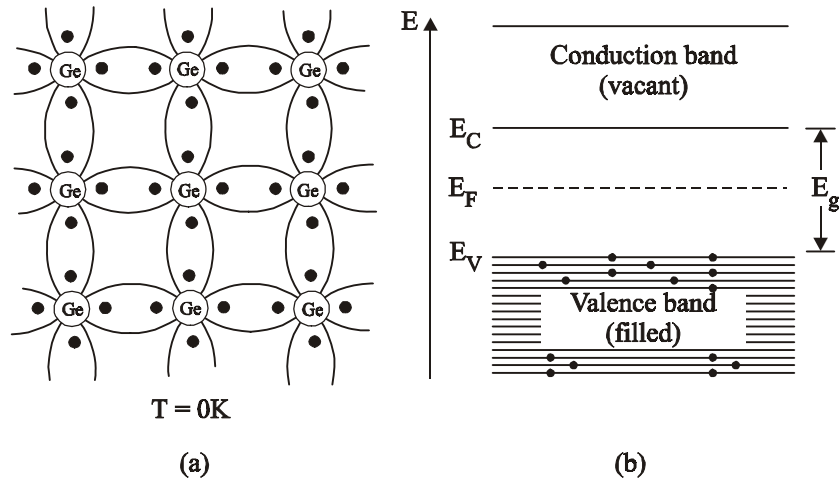
*A semiconductor in extremely pure form is known as intrinsic semiconductor. Its electrical conductivity is changed only by thermal excitation.*

The common examples for intrinsic semiconductors are pure **silicon (Si)** and **germanium (Ge)**. They belong to fourth group elements in the periodic table. Germanium has 32 electrons and silicon has 14 electrons in their atomic structures.

They are **tetravalent atoms** since they have four valence electrons. The neighbouring atoms form **covalent bonds** by sharing four electrons with each other so as to form a **stable structure**.

## 2.3 ENERGY BAND DIAGRAM

Fig. 2.3 shows a two-dimensional crystal structure of germanium and energy band representation of intrinsic semiconductor at very low temperature.



**Fig. 2.3 Germanium crystal at 0K**

**Fig. 2.3(a) Two-dimensional representation of germanium solid. No free electron is available as all the valence electrons are engaged in covalent bonds.**

**Fig. 2.3(b) Energy band representation. Valence band is fully occupied and conduction band is completely vacant.**

At very low temperature say 0K, no free electrons are available for conduction. Hence, this semiconductor behaves as an insulator at very low temperature.

### **Charge carriers in intrinsic semiconductor**

To get free electrons, covalent bonds must be broken. There are many ways of breaking covalent bond and setting the electrons free. One such way is to increase crystal temperature above 0K.

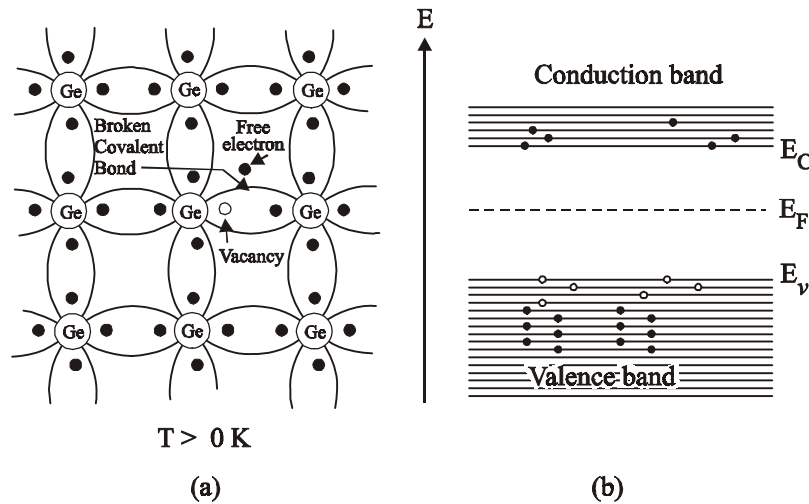
When the temperature of intrinsic semiconductor is increased, some of the electrons get sufficient energy to break covalent bonds.

**Once the electrons are liberated from bond, they become free electrons. These free electrons move randomly through crystal. (Fig. 2.4(a))**

As shown in fig. 2.4 (b), the energy required to break a covalent bond and to set an electron free is equal to band gap energy  $E_g$ . It is about 0.72 eV for germanium and 1.1 eV for silicon.

When an electron acquires energy  $E_g$ , it jumps from valence band to conduction band. As a result, a vacant site (empty space) is created in valence band.

**This vacant site is called as a hole. The absence of an electron in covalent bond is known as hole. A hole can attract an electron and hence it acts as a positive charge.**



**Fig. 2.4 Germanium crystal at temperature above 0K.**

- (a) **Thermal vibrations of atoms lead to breaking up of covalent bonds. Consequently, a free electron and a vacancy are produced simultaneously.**
- (b) **Energy band representation. Energy  $E_g (=E_c - E_v)$  causes transition of electrons from valence band to conduction band, leaving vacancies (hole) behind.**

When an electrical field is applied, these free electrons acquire directional motion and contribute to electrical conductivity.

For every electron freed from covalent bond, one hole is created in the crystal. It is relatively easy for a valence electron in a neighbouring atom to leave its covalent bond and fill this hole.

As a result, an electron moving from a covalent bond to fill a hole leaves behind a hole in its original position.

The hole effectively moves in a direction opposite to that of an electron. The hole in its new position may now be filled by an electron from another covalent bond.

Thus hole will correspondingly move one more step in the direction opposite to the motion of the electron.

Therefore, in intrinsic semiconductor, current conduction is due to the movement of both electrons and holes.

Here, the number of electrons is equal to the number of holes at any given temperature.

## 2.4

### CARRIER CONCENTRATION IN INTRINSIC SEMICONDUCTORS

#### Definition

The number of electrons in conduction band per unit volume of the material is called as electron concentration ( $n$ ).

Similarly the number of holes in valence band per unit volume of the material is called hole concentration ( $p$ ).

**In general, the number of charge carriers per unit volume of the material is called carrier concentration. It is also known as density of charge carriers.**

#### Density of Electrons in Conduction Band (Derivation)

The number of electrons per unit volume in conduction band for energy between  $E$  and  $E + dE$  is given by

$$dn = Z(E) F(E) dE \quad \dots(1)$$

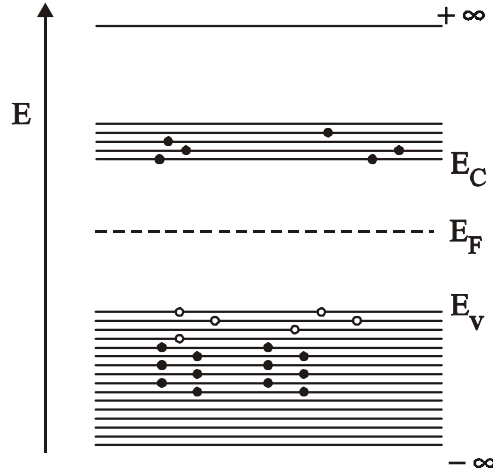
where  $Z(E) dE$  – Density of states in energy between  $E$  and  $E + dE$

$F(E)$  – Probability of electron occupancy.

Number of electrons in conduction band for the entire range is calculated by integrating eqn (1) between energy  $E_C$  and  $+\infty$ .

$$\int dn = n = \int_{E_C}^{+\infty} Z(E) F(E) dE \quad \dots(2)$$

$E_C$  is energy corresponding to the bottom most level and  $+\infty$  is energy corresponding to the upper most level in conduction band. (Fig 2.5).



**Fig. 2.5 Energy band diagram of intrinsic semiconductor**

Density of states in conduction band between the energy range  $E$  and  $E + dE$  is given by

$$Z(E) dE = \frac{4\pi}{h^3} (2m_e^*)^{3/2} E^{1/2} dE \quad \dots(3)$$

The bottom edge of the conduction band ( $E_C$ ) denotes the potential energy of an electron at rest. Therefore,  $(E - E_C)$  is the kinetic energy of conduction electron at higher energy levels.

Thus, in eqn (3),  $E$  is replaced as  $(E - E_C)$

$$Z(E) dE = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_C)^{1/2} dE \quad \dots(4)$$

The electrons in conduction band are not totally free. They move in a periodic potential of the crystal lattice. Therefore, in eqn (3), the mass of the electron ( $m$ ) is replaced by its effective mass  $m_e^*$  according to band theory of solids.

The probability of electron occupation is given by Fermi distribution function

$$F(E) = \frac{1}{1 + e^{(E - E_F)/kT}} \quad \dots(5)$$

Substituting eqns (4) and (5) in (2), we get

$$n = \int_{E_C}^{+\infty} \frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_C)^{1/2} \times \frac{1}{1 + e^{(E - E_F)/kT}} dE$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{+\infty} \frac{(E - E_C)^{1/2}}{1 + e^{(E - E_F)/kT}} dE \dots(6)$$

Since  $kT$  is very small and  $(E - E_F)$  is greater than  $kT$ ,  $e^{(E - E_F)/kT}$  is very large compared to '1' Hence, '1' from the denominator of eqn (6) is neglected.

$$\text{ie., } 1 + e^{(E - E_F)/kT} \approx e^{(E - E_F)/kT}.$$

Now, eqn (6) becomes

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{+\infty} \frac{(E - E_C)^{1/2} dE}{e^{(E - E_F)/kT}}$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{+\infty} (E - E_C)^{1/2} e^{-(E - E_F)/kT} dE$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{+\infty} (E - E_C)^{1/2} e^{(E_F - E)/kT} dE$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{E_F/kT} \int_{E_C}^{\infty} (E - E_C)^{1/2} e^{-E/kT} dE \quad \dots(7)$$

To evaluate above integral in eqn (7), let us assume

$$\begin{array}{l|l|l} E - E_C = x & \text{when} & \text{when} \\ E = E_C + x & E = E_C & E = +\infty \\ dE = dx & E_C - E_C = x & +\infty - E_C = x \\ & \therefore x = 0 & \therefore x = +\infty \end{array}$$

Substituting above values in eqn (7), we have

$$\begin{aligned} n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{E_F/kT} \int_0^\infty x^{1/2} e^{-(E_C+x)/kT} dx \\ n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{(E_F-E_C)/kT} \int_0^\infty x^{1/2} e^{-x/kT} dx \quad \dots (8) \end{aligned}$$

Using the gamma function, it is shown that

$$\int_0^\infty x^{1/2} e^{-x/kT} dx = \frac{(kT)^{3/2} \pi^{1/2}}{2} \quad \dots (9)$$

Substituting eqn (9) in eqn (8), we have

$$\begin{aligned} n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{(E_F-E_C)/kT} \left[ \frac{(kT)^{3/2} \pi^{1/2}}{2} \right] \\ n &= \frac{2\pi}{h^3} (2m_e^*)^{3/2} (kT)^{3/2} \pi^{1/2} e^{(E_F-E_C)/kT} \\ n &= \frac{2\pi\pi^{1/2} (2m_e^*)^{3/2} (kT)^{3/2} e^{(E_F-E_C)/kT}}{(h^2)^{3/2}} \\ \boxed{n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_F-E_C)/kT}} & \quad \dots(10) \end{aligned}$$

Equation (10) is the expression for concentration of electrons in the conduction band of intrinsic semiconductor.

### Density of holes in Valence Band of Intrinsic Semiconductor (Derivation)

We know that if an electron is transferred from valence band to conduction band, a hole is created in valence band.

Let  $dp$  be the number of holes per unit volume in valence band between the energy  $E$  and  $E + dE$ .

$$dp = Z(E) (1 - F(E)) dE \quad \dots(1)$$

where  $Z(E) dE \rightarrow$  Density of states in the energy range  $E$  and  $E + dE$ .

Since  $F(E)$  is the probability of electron occupation  $1 - F(E)$  is the probability of an unoccupied electron state, i.e., probability of presence of hole.

$$\begin{aligned} 1 - F(E) &= 1 - \frac{1}{1 + e^{(E - E_F)/kT}} \\ &= \frac{1 + e^{(E - E_F)/kT} - 1}{1 + e^{(E - E_F)/kT}} \\ 1 - F(E) &= \frac{e^{(E - E_F)/kT}}{1 + e^{(E - E_F)/kT}} \quad \dots (2) \end{aligned}$$

Since  $E$  is very small when compared to  $E_F$  in valence band,  $(E - E_F)$  is a negative quantity. Therefore,  $e^{(E - E_F)/kT}$  is very small and it is neglected in the denominator term of eqn (2).

$$\begin{aligned} \text{i.e.,} \quad 1 + e^{(E - E_F)/kT} &\approx 1 \\ \therefore 1 - F(E) &= e^{(E - E_F)/kT} \quad \dots(3) \end{aligned}$$

Density of states in valence band,

$$Z(E) dE = \frac{4\pi}{h^3} (2m_h^*)^{3/2} E^{1/2} dE \quad \dots(4)$$

Here,  $m_h^*$  is the effective mass of the hole in valence band.

$E_v$ , top of energy level in valence band is the potential energy of a hole at rest. Hence,  $(E_v - E)$  is the kinetic energy of the hole at level below  $E_v$ . So the term  $E$  in eqn (4) is replaced as  $(E_v - E)$ .

$$Z(E) dE = \frac{4\pi}{h^3} (2m_h^*)^{3/2} (E_v - E)^{1/2} dE \quad \dots(5)$$

Substituting eqns (3) and (5) in (1), we get

$$dp = \frac{4\pi}{h^3} (2m_h^*)^{3/2} (E_v - E) e^{(E - E_F)/kT} dE \quad \dots(6)$$

The number of holes in valence band for the entire energy range is obtained by integrating eqn (6) between limits  $-\infty$  to  $E_v$ .

$$\int_{-\infty}^{E_v} dp = p = \int_{-\infty}^{E_v} \frac{4\pi}{h^3} (2m_h^*)^{3/2} (E_v - E)^{1/2} e^{(E - E_F)/kT} dE$$

$$p = \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(-E_F/kT)} \int_{-\infty}^{E_v} (E_v - E)^{1/2} e^{E/kT} dE \quad \dots(7)$$

To evaluate the integral in eqn (7), let us assume,

|                                                                           |                                                                                                                      |                                                                                  |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| $\begin{aligned} E_v - E &= x \\ E &= E_v - x \\ dE &= -dx \end{aligned}$ | <p>when</p> $\begin{aligned} E &= -\infty \\ E_v - (-\infty) &= x \\ E_v + \infty &= x \\ x &= \infty \end{aligned}$ | <p>when</p> $\begin{aligned} E &= E_v \\ E_v - E_v &= x \\ x &= 0 \end{aligned}$ |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|

Substituting these values in eqn (7), we have

$$p = \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(-E_F/kT)} \int_{-\infty}^0 x^{1/2} e^{(E_v - x)/kT} (-dx) \quad \dots(8)$$

$$p = \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(E_v - E_F)/kT} \int_0^{\infty} x^{1/2} e^{-x/kT} dx \quad \dots(9)$$

[ -ve sign is omitted by interchanging the limits]

Using the gamma function, it is shown that

$$\int_0^{\infty} x^{1/2} e^{-x/kT} dx = \frac{(kT)^{3/2} \pi^{1/2}}{2} \quad \dots (10)$$

Substituting eqn (10) in eqn (9), we have

$$p = \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(E_v - E_F)/kT} \left[ \frac{(kT)^{3/2} \pi^{1/2}}{2} \right]$$

$$p = \frac{2\pi}{h^3} (2m_h^*)^{3/2} (kT)^{3/2} \pi^{1/2} e^{(E_v - E_F)/kT}$$

$$p = 2\pi \pi^{1/2} \frac{(2m_h^*)^{3/2} (kT)^{3/2} e^{(E_v - E_F)/kT}}{(h^2)^{3/2}}$$

$$p = 2 \left( \frac{2 \pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_F) / kT} \quad \dots(11)$$

The equation (11) is the expression for the concentration of holes in valence band of intrinsic semiconductor.

### INTRINSIC CARRIER CONCENTRATION

In an intrinsic semiconductor, the number of electrons in conduction band is equal to the number of holes in valence band.

In general, intrinsic carrier concentration  $n_i$  is equal to electrons concentration in conduction band ( $n$ ) or holes concentration in valence band ( $p$ ).

$$\text{i.e.,} \quad n_i = n = p \quad \dots(1)$$

$$n_i \times n_i = n_i^2 = np \quad \dots(2)$$

Substituting the expressions of  $n$  and  $p$  in eqn (2), we have

$$\begin{aligned} n_i^2 &= 2 \left( \frac{2 \pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_F - E_C) / kT} \times 2 \left( \frac{2 \pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_F) / kT} \\ n_i^2 &= 4 \left( \frac{2 \pi kT}{h^2} \right)^3 (m_e^* m_h^*)^{3/2} e^{(E_v - E_C) / kT} \\ n_i^2 &= 4 \left( \frac{2 \pi kT}{h^2} \right)^3 (m_e^* m_h^*)^{3/2} e^{-E_g / kT} \end{aligned} \quad \dots(3)$$

where  $E_C - E_v = E_g$  is forbidden energy gap.

Taking square root on both sides in eqn (3), we have

$$(n_i^2)^{1/2} = \left( 4 \left( \frac{2\pi kT}{h^2} \right)^3 (m_e^* m_h^*)^{3/2} e^{-E_g/kT} \right)^{1/2}$$

$$n_i = 4^{1/2} \left( \frac{2\pi kT}{h^2} \right)^{3/2} ((m_e^* m_h^*)^{3/2})^{1/2} (e^{-E_g/kT})^{1/2}$$

$$n_i = 2 \left( \frac{2\pi kT}{h^2} \right)^{3/2} (m_e^* m_h^*)^{3/4} e^{-E_g/2kT}$$

...(4)

The eqn (4) is expression for intrinsic carrier concentration

### Fermi Level and Its Variation with temperature

**Fermi level is a characteristic energy level of the material. The position of Fermi level is important in determining the electron and hole concentrations in a semiconductor.**

**In intrinsic semiconductor, the number of electrons in conduction band is equal to the number of holes in valence band.**

$$\text{i.e.,} \quad n = p \quad \dots(1)$$

Substituting the expressions of  $n$  and  $p$  in eqn (1), we have

$$2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_F - E_C)/kT} = 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_F)/kT}$$

$$(m_e^*)^{3/2} e^{(E_F - E_C)/kT} = (m_h^*)^{3/2} e^{(E_v - E_F)/kT}$$

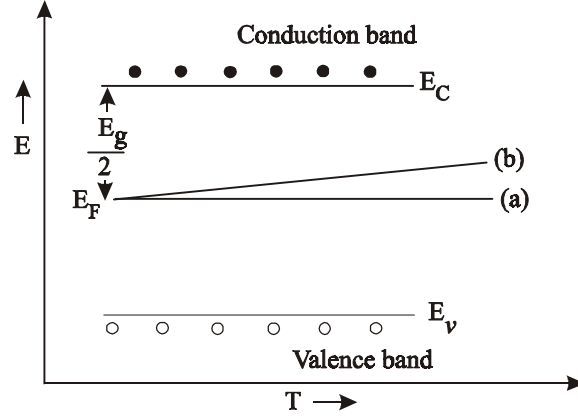
Rearranging, we get

$$E_F = \frac{E_v + E_C}{2} + \frac{3kT}{4} \log_e \left( \frac{m_h^*}{m_e^*} \right)$$

...(2)

$$\text{If } m_h^* = m_e^*, \text{ then } \log_e \left( \frac{m_h^*}{m_h^*} \right) = \log_e 1 = 0.$$

Hence, eqn (2) becomes



**Fig. 2.6** Positions of Fermi level in intrinsic semiconductor at various temperatures.

- (a) At  $T = 0K$ , Fermi level is in the middle of the forbidden band.
- (b) As the temperature rises, it shifts upward since  $m_h^* > m_e^*$ .

$$\boxed{E_F = \frac{E_v + E_C}{2}} \quad \dots (3)$$

Thus, Fermi level is located half way between the top of valence band and bottom of conduction band as shown in fig.2.6 (a). Its position is independent of temperature.

In reality  $m_h^* > m_e^*$ , Fermi level is just above the middle of energy gap and it rises slightly with increasing temperature.

### Limitations of intrinsic semiconductor

Intrinsic semiconductors cannot be directly used to fabricate devices due to the following limitations:

- Electrical conductivity is low. Germanium has a conductivity of  $1.67 \Omega^{-1} \text{m}^{-1}$  which is nearly  $10^7$  times smaller than that of copper.
- Electrical conductivity is a function of temperature and increases exponentially as temperature increases.

In intrinsic or pure semiconductors, the carrier concentration of both electrons and holes is very low at normal temperatures.

In order to get sufficient current density through semiconductor, a large electrical field should be applied. This problem is overcome by adding suitable impurities into intrinsic semiconductors.

## 2.5 EXTRINSIC OR IMPURE SEMICONDUCTORS

**In a semiconducting material, if the charge carriers originate from impurity atoms which are doped to the original material, then this type of semiconductor is known as extrinsic or impure semiconductor.**

**It is also known as doped semiconductor.**

Extrinsic semiconductor is obtained by adding trivalent or pentavalent impurities atoms to a tetravalent semiconductor. The electrical properties of pure semiconductors can be easily changed even with the addition of **very little amount of impurities**.

### ***Doping***

*The addition of impurities to a pure semiconductor is known as **doping** and added impurity is called as **doping agent or dopant**.*

The addition of impurities increases the number of free electrons and holes in semiconductor and hence increases its electrical conductivity.

Some of the common doping agents are arsenic, antimony, phosphorus, gallium, aluminium and boron. These elements have either five or three valence electrons in the outermost orbit.

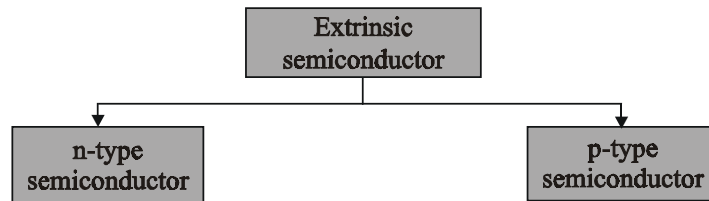
**Advantages of Extrinsic semiconductors**

- Electrical conductivity is high.
- Electrical conductivity can be altered to any desired value by controlling of doping concentration.
- Electrical conductivity is not a function of temperature.

**Types of Extrinsic semiconductors**

The extrinsic semiconductors are classified into two types based on the type of impurity added.

- n - type semiconductor*
- p - type semiconductor*

**n - type semiconductor**

When a small amount of pentavalent impurity (group V element) is doped to a pure semiconductor, it becomes **n - type semiconductor**.

Such impurities are known as **donor** impurities because they **donate** free electrons to semiconductor crystal.

Typical examples of pentavalent impurities are phosphorus, (Atomic No. 15) and antimony (Atomic No. 51).

**Covalent bond in n - type semiconductor**

A pentavalent impurity (phosphorus) having five valence electrons is added to a pure semiconductor having four valence electrons (silicon or germanium).

Now, four electrons of germanium form a **covalent bond** with four valence electrons of phosphorus (impurity atom).

The fifth electron which is now free finds no place in covalent bond structure as shown in fig. 2.7 (a).

We have one electron left free. This acts as a conduction electron. A very small amount of energy (0.01 eV for germanium and 0.05 eV for silicon) is needed to detach this fifth electron.

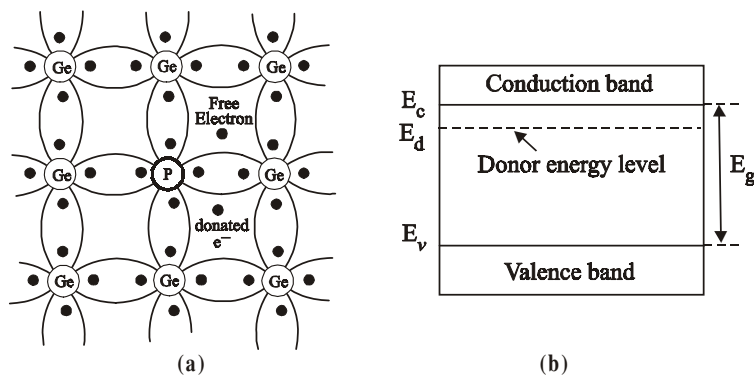
**The addition of pentavalent impurity gives a large number of free electrons (negative charges) in semiconductor. Therefore, it is called  $n$  - type semiconductor where  $n$  stands for negative type.**

Since every pentavalent atom contributes one free electron, in addition to thermally generated electron-hole pairs, the number of free electrons is more than the number of holes in  $n$ -type semiconductor.

Thus in this case, **electrons are majority charge carriers and holes are minority charge carriers.**

### Energy band of $n$ - type semiconductor

The energy band diagram of  $n$  - type semiconductor is shown in fig.2.6 (b). When the donor impurities are added, the allowable energy levels (donor energy levels) are introduced.



**Fig. 2.7  $n$ -type semiconductor**

**(a) crystal structure (b) Energy band diagram**

These donor energy levels are slightly below the conduction band. They are discrete and do not form a band because the impurity atoms are far away in the crystal and hence their interaction is small.

The donor energy level for germanium is 0.01 eV and for silicon it is 0.05 eV below the conduction band. Therefore, even at room temperature, almost all the fifth electrons enter into the conduction band.

## 2.6 CARRIER CONCENTRATION IN $n$ - TYPE SEMICONDUCTORS [Derivation]

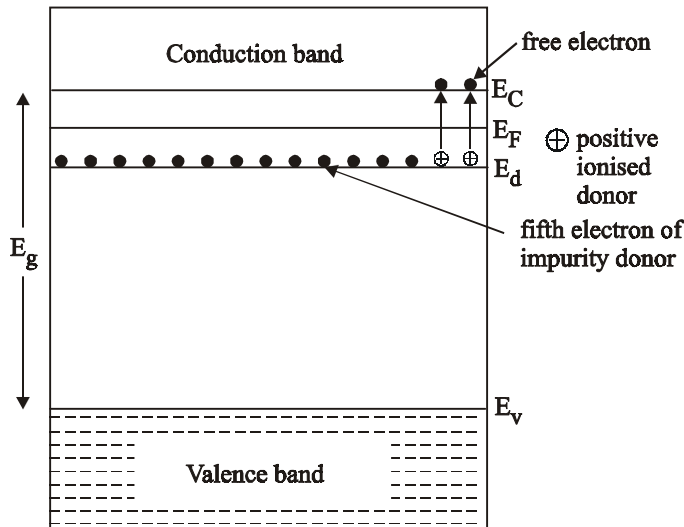
The energy band diagram of  $n$  - type semiconductor is shown in figure 2.8. In  $n$  - type semiconductor, the donor level is just below conduction band.

Density of electrons per unit volume in conduction band is given by

$$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_F - E_C) / kT} \quad \dots(1)$$

$E_C$  – Energy corresponding to the bottom most level of conduction band.

Density of ionised donors =  $N_d [1 - F(E_d)]$



**Fig. 2.8 Energy band diagram of  $n$ -type semiconductor**

Since,  $F(E_d)$  is the probability for finding electron in donor energy level (unionised donor), therefore  $1 - F(E_d)$  is the probability for finding ionised donors.

$E_d$  represents the donor energy level and  $N_d$  denotes donor concentration i.e., the number of donor atoms per unit volume of the material.

$$= N_d \left[ 1 - \frac{1}{1 + e^{(E_d - E_F)/kT}} \right] \quad \dots(2)$$

$$= N_d \left[ \frac{1 + e^{(E_d - E_F)/kT} - 1}{1 + e^{(E_d - E_F)/kT}} \right]$$

$$= \frac{N_d e^{(E_d - E_F)/kT}}{1 + e^{(E_d - E_F)/kT}} \quad \dots(3)$$

$e^{(E_d - E_F)/kT}$  is very small in eqn (3) when compared to '1'. Hence, it is neglected.

$$\therefore 1 + e^{(E_d - E_F)/kT} \approx 1$$

$$\text{Density of ionised donor} = N_d e^{(E_d - E_F)/kT} \quad \dots(4)$$

At equilibrium, the density of electron in conduction band is equal to the density of ionised donors.

Equating (1) and (4), we get

$$2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_F - E_C)/kT} = N_d e^{(E_d - E_F)/kT} \quad \dots(5)$$

rearranging the terms, we have,

$$\frac{e^{(E_F - E_C)/kT}}{e^{(E_d - E_F)/kT}} = \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}}$$

$$e^{(E_F - E_C)/kT} e^{-(E_d - E_F)/kT} = \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}}$$

$$e^{(E_F - E_C - E_d + E_F)/kT} = \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \quad \dots(6)$$

Taking log on both sides, we have

$$\log_e e^{(E_F - E_C - E_d + E_F)/kT} = \log_e \left[ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right]$$

$$\frac{E_F - E_C - E_d + E_F}{kT} = \log_e \left[ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right]$$

$$\left[ \cdot \cdot \log_e e^x = x \right]$$

$$2E_F - (E_C + E_d) = kT \log_e \left[ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right]$$

$$\text{or} \quad 2E_F = E_d + E_C + kT \log_e \left[ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right]$$

$$E_F = \frac{E_d + E_C}{2} + \frac{kT}{2} \log_e \left[ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right] \quad \dots(7)$$

Substituting the expression of  $E_F$  from (7) in (1), we get

$$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} \exp \left[ \frac{\left( \frac{E_d + E_C}{2} + \frac{kT}{2} \log_e \left\{ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right\} - E_C \right)}{kT} \right] \quad \dots (8)$$

$$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} \exp \left[ \frac{E_d + E_C - 2E_C}{2kT} + \frac{1}{2} \log_e \left\{ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right\} \right]$$

$$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} \exp \left[ \frac{E_d - E_C}{2kT} + \log_e \left\{ \frac{N_d^{1/2}}{\left\{ 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} \right\}^{1/2}} \right\} \right]$$

$$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_d - E_C)/2kT} \cdot \log_e \left\{ \frac{N_d^{1/2}}{\left\{ 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} \right\}^{1/2}} \right\}$$

$$n = 2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_d - E_C)/2kT} \frac{N_d^{1/2}}{2^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/4}} \quad \dots (9)$$

Rearranging the expression (9), we get

$$\begin{aligned}
 n &= 2 \frac{N_d^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}}{2^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/4}} e^{(E_d - E_C)/2kT} \\
 n &= 2^{1/2} \times 2^{1/2} \frac{N_d^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}}{2^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/4}} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{-3/4} e^{(E_d - E_C)/2kT} \\
 n &= 2^{1/2} N_d^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/4} e^{(E_d - E_C)/2kT} \quad \dots (10)
 \end{aligned}$$

$$\boxed{n = (2N_d)^{1/2} \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/4} e^{-\Delta E / 2kT}} \quad \dots(11)$$

where  $\Delta E = E_C - E_d$  is the ionisation energy of the donor. i.e.,  $\Delta E$  denotes the amount of energy required to transfer an electron from donor energy level  $E_d$  to conduction band  $E_C$ .

## Results

- The density of electrons in conduction band is proportional to the square root of the donor concentration. The equation (11) is valid only at low temperatures.
- At high temperature, we must take into account of intrinsic carrier concentration of semiconductor due to breaking of covalent bond along with electron concentration produced by donor impurity.

- At very high temperatures, intrinsic carrier concentration which is generated thermally due to breaking of covalent bond over takes electrons due to donor impurity.
- That is, at very high temperature, n - type semiconductor behaves like intrinsic semiconductor and donor concentration becomes insignificant.

### Fermi level

Fermi level gives the probability of finding an electron at a given energy value. If Fermi level lies exactly at the middle of the two levels, then the probability of finding an electron is half, e.g., as in an intrinsic semiconductor.

**In extrinsic semiconductor, Fermi level strongly depends on temperature as well as the nature of doping and doping concentration.**

The Fermi level is little below conduction band in n-type semiconductor and it is just above valence band in p-type semiconductor.

### 2.7

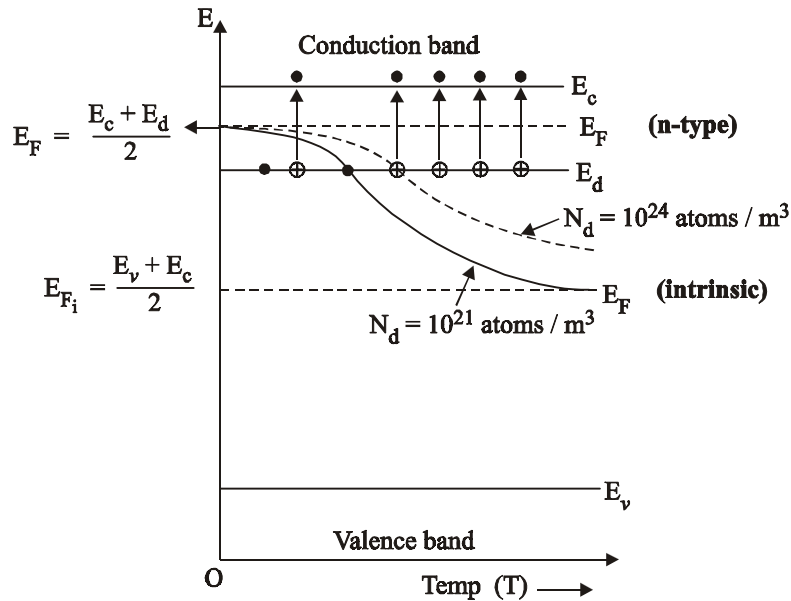
#### VARIATION OF FERMİ LEVEL WITH TEMPERATURE AND IMPURITY CONCENTRATION (n - type semiconductor)

Fermi level in *n* - type semiconductor is given by

$$E_F = \frac{E_d + E_c}{2} + \frac{kT}{2} \log_e \left[ \frac{N_d}{2 \left( \frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right] \dots (1)$$

At  $T = 0K$ , the eqn (1) reduces to

$$E_F = \frac{E_d + E_c}{2}$$



**Fig. 2.9 Variation of Fermi level with temperature and impurity concentration in n-type semiconductor**

- At 0K, Fermi level lies exactly in the middle of the donor level ( $E_d$ ) and bottom of the conduction band as shown in fig. 2.9.
- When temperature is gradually increased from a low temperature, the contribution of electron to conduction band from valence band increases.
- At very high temperature (500 K in germanium), concentration of electron in conduction band due to breaking of covalent bond i.e., from valence band exceeds very much donor concentration and hence intrinsic behaviour predominates at higher temperature.
- Fermi level shifts downwards when temperature is increased and finally reaches the middle of the band gap (or) that of intrinsic Fermi level.
- When the concentration of donors increases, the extrinsic behaviour increases.

### ***p*-type semiconductor**

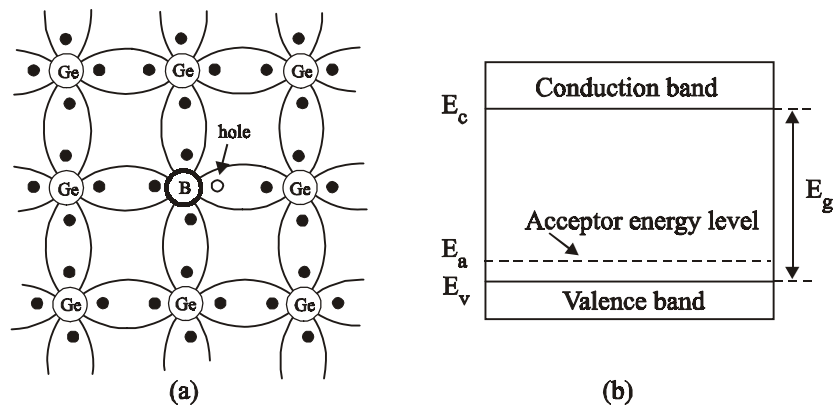
**When a small amount of trivalent impurity is doped to a pure semiconductor, it becomes *p* - type semiconductor.**

The addition of trivalent impurity provides a large number of holes in semiconductor.

Typical examples of trivalent impurities are gallium (Atomic No: 31) and indium (Atomic No. 49). Such impurities are known as **acceptor impurities** because holes created can **accept electrons**.

In a pure semiconductor (germanium) having 4 valence electrons, if a trivalent impurity (boron) having '3' valence electrons is added, then 3 valence electrons of trivalent impurity form a covalent bond with three valence electrons of germanium.

The fourth valence electron of Ge atom is unable to form a covalent bond. The incomplete covalent bond is being short of one **electron**. This **missing electron** is called a **hole**. (Fig. 2.10(a))



**Fig. 2.10 *p* - type semiconductor (a) crystal structure  
(b) energy band diagram**

Every trivalent impurity atom contributes one hole in addition to thermally generated electron - hole pairs. Therefore, number of holes is more than number of electrons.

The addition of trivalent impurity creates large number of holes (positive charge carriers) in semiconductor and hence it is called *p*-type semiconductor where *p* stands for positive type.

Hence in this type of semiconductor, holes are majority charge carriers and electrons are minority charge carriers.

In this case, allowable energy level (acceptor energy level) is created just above valence band (fig. 2.10(b)).

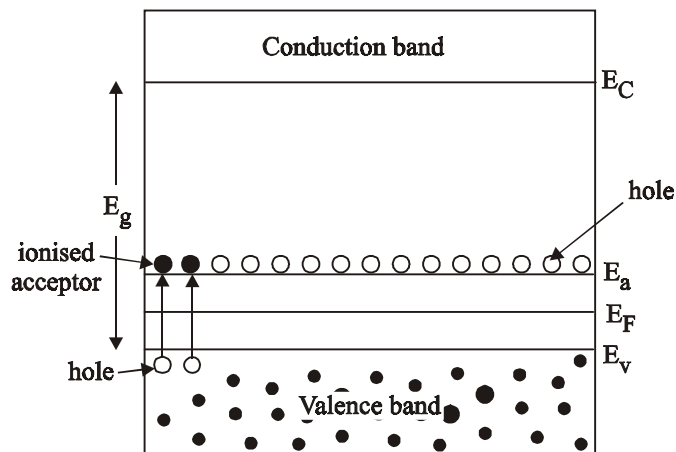
A very small amount of energy is needed for an electron to enter acceptor energy level from valence band. Thus, a hole is generated in the valence band corresponding to each ionised acceptor.

In other words, a large number of positive charge carriers are created.

## 2.8

### CONCENTRATION OF HOLES IN VALENCE BAND OF *p* – TYPE SEMICONDUCTORS [Derivation]

In *p*-type semiconductor, acceptor energy level is just above valence band (fig. 2.11).



**Fig. 2.11** Energy band diagram for *p*-type semiconductor

Density of holes per unit volume in valence band is given by

$$p = 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_F)/kT} \quad \dots(1)$$

$E_v \rightarrow$  Energy corresponding to top most level of valence band.

$$\text{Density of ionised acceptors} = N_a F(E_a) \quad \dots (1)$$

$N_a$  – the number of acceptor atoms per unit volume.

$$F(E_a) = \frac{1}{1 + e^{(E_a - E_F)/kT}}$$

$E_a$  – acceptor energy level

Here,  $F(E_a)$  is probability for finding electron in acceptor energy level ie., ionised acceptor.

The eqn (1) becomes, density of ionised acceptors

$$= \frac{N_a}{1 + e^{(E_a - E_F)/kT}} \quad \dots(2)$$

$e^{(E_a - E_F)/kT}$  is a large quantity and thus ‘1’ from the denominator of R.H.S. of eqn (2) is neglected.

Now, the eqn (2) is modified as,

$$N_a F(E_a) = \frac{N_a}{e^{(E_a - E_F)/kT}}$$

$$N_a F(E_a) = N_a e^{-(E_a - E_F)/kT}$$

$$\text{Density of ionised acceptors} = N_a e^{(E_F - E_a)/kT} \quad \dots(3)$$

At equilibrium,

$$\left( \begin{array}{c} \text{Density of holes} \\ \text{in valence band} \end{array} \right) = \left( \begin{array}{c} \text{Density of} \\ \text{ionised acceptors} \end{array} \right)$$

$$2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_F)/kT} = N_a e^{(E_F - E_a)/kT} \quad \dots(4)$$

rearranging eqn (4), we have

$$\frac{e^{(E_v - E_F)/kT}}{e^{(E_F - E_a)/kT}} = \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}}$$

$$e^{(E_v - E_F)/kT} \cdot e^{-(E_F - E_a)/kT} = \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}}$$

$$e^{(E_v - E_F - E_F + E_a)/kT} = \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \quad \dots (5)$$

Taking log on both sides in eqn (5), we have

$$\log_e e^{(E_v - E_F - E_F + E_a)/kT} = \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}}$$

$$\frac{E_v - 2E_F + E_a}{kT} = \log_e \left[ \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right]$$

$$E_a + E_v - 2E_F = kT \log_e \left[ \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right] \quad \dots (6)$$

Rearranging,

$$2E_F = E_a + E_v - kT \log_e \left[ \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right]$$

$$E_F = \frac{E_a + E_v}{2} - \frac{kT}{2} \log_e \left[ \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right]$$

... (7)

Substituting eqn of  $E_F$  from (7) in (1), we get

$$p = 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} \exp \left[ E_v - \left( \frac{E_a + E_v}{2} \right) - \frac{kT}{2} \log_e \left[ \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right] \right]$$

$$p = 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} \exp \left[ \frac{E_v - \left( \frac{E_v - E_a}{2} \right) + \frac{kT}{2} \log_e \left[ \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right]}{kT} \right] \quad \dots(8)$$

$$p = 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} \exp \left[ \frac{2E_v - E_v - E_a}{2kT} + \frac{1}{2} \log_e \left[ \frac{N_a}{2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right] \right]$$

$$\begin{aligned}
p &= 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} \exp \left[ \frac{E_v - E_a}{2kT} + \log_e \left[ \frac{\left( \frac{N_a}{2} \right)^{1/2}}{\left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right] \right] \\
p &= 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} \exp \left[ \frac{E_v - E_a}{2kT} + \log_e \left[ \frac{\left( \frac{N_a}{2} \right)^{1/2}}{\left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right] \right] \\
p &= 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_a) / 2kT} e^{\log_e \left[ \frac{\left( \frac{N_a}{2} \right)^{1/2}}{\left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/4}} \right]} \\
p &= 2 \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_a) / 2kT} \frac{\left( \frac{N_a}{2} \right)^{1/2}}{\left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/4}} \\
p &= 2 \frac{N_a^{1/2}}{2^{1/2}} \frac{\left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2}}{\left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/4}} e^{(E_v - E_a) / 2kT} \\
p &= \frac{2^{1/2} 2^{1/2} N_a^{1/2}}{2^{1/2}} \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/2} \left( \frac{2\pi m_h^* kT}{h^2} \right)^{-3/4} e^{(E_v - E_a) / 2kT}
\end{aligned}$$

$$p = (2 N_a)^{1/2} \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/4} e^{(E_v - E_a)/2kT} \quad \dots(9)$$

$$p = (2 N_a)^{1/2} \left( \frac{2\pi m_h^* kT}{h^2} \right)^{3/4} e^{-\Delta E/2kT} \quad \dots(10)$$

where  $\Delta E = E_a - E_v$  is the ionisation energy of acceptors.

### Results

- Density of holes in valence band is proportional to the square root of acceptor concentration.
- At high temperature, we must take into account of the intrinsic carrier concentration of semiconductor due to breaking of covalent bond along hole concentration produced by acceptor impurity.
- At very high temperature, intrinsic carrier concentration over takes holes due to acceptor concentration.
- i.e., At very high temperature,  $p$  - type semiconductor behaves like an intrinsic semiconductor and acceptor concentration becomes insignificant.

## 2.9

### VARIATION OF FERMI LEVEL WITH TEMPERATURE AND IMPURITY CONCENTRATION IN $p$ - TYPE SEMICONDUCTOR

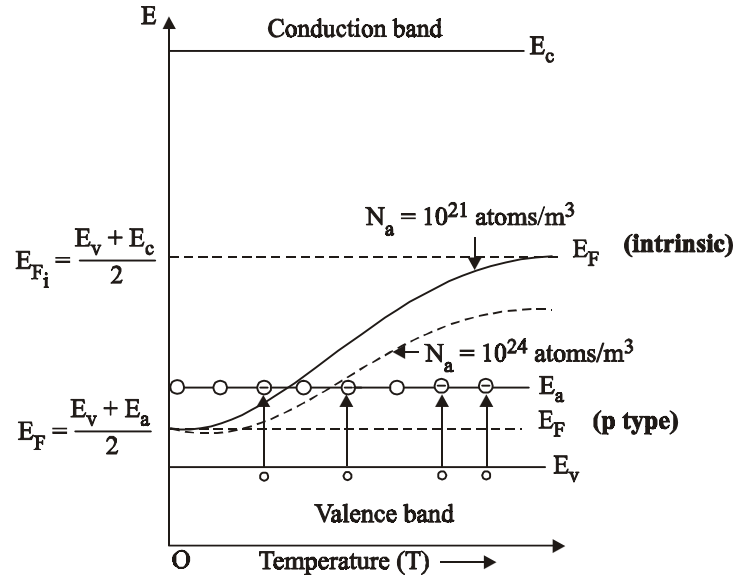
Fermi level in  $p$  - type semiconductor is given by

$$E_F = \frac{E_a + E_v}{2} - \frac{kT}{2} \log_e \left[ \frac{N_a}{2 \left[ \frac{2\pi m_h^* kT}{h^2} \right]^{3/2}} \right] \quad \dots (1)$$

Substituting  $T = 0 \text{ K}$  in expression (1), we have

$$E_F = \frac{E_a + E_v}{2} \quad \dots (2)$$

- At 0 K, Fermi level lies exactly middle of the acceptor level  $E_a$  and top of valence band  $E_v$ . (Fig 2.12)



**Fig. 2.12** Variation of Fermi level with temperature and impurity concentration in p-type semiconductor

- As temperature increases, more and more acceptor atoms are ionised. Fermi level  $E_F$  (p-type) shifts upwards. At a particular temperature, when all the acceptor atoms are ionised, Fermi level crosses acceptor level.
- At very high temperature, Fermi level shifts to intrinsic Fermi level and p-type semiconductor behaves like an intrinsic semiconductor.
- Further, when the concentration of acceptor atoms increases, extrinsic behaviour increases. Fermi level reaches intrinsic Fermi level at very high temperature.

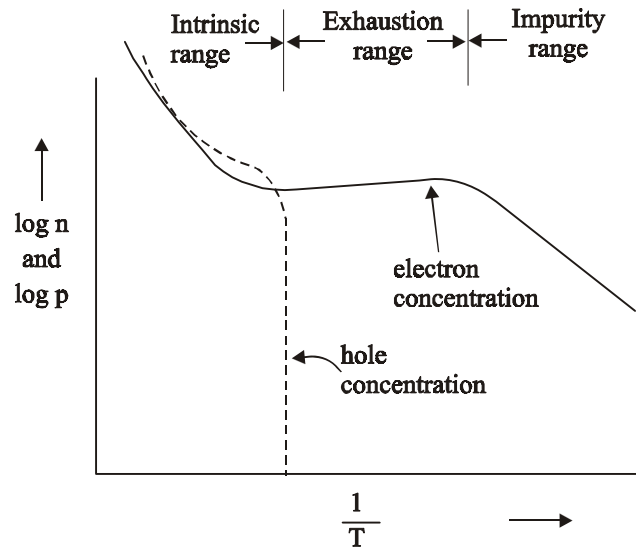
## 2.10

### VARIATION OF CARRIER CONCENTRATION WITH TEMPERATURE AND IMPURITY

In extrinsic semiconductor, the resistivity decreases linearly with increase in temperature. This variation is considered under three different regions.

- i) ***Extrinsic or impurity range***
- ii) ***Exhaustion range***
- iii) ***Intrinsic range***

For a  $n$ -type semiconductor, the variation of carrier concentration  $n$  and  $p$  with temperature is shown in fig. 2.13.



**Fig. 2.13** Variation of carrier concentration with temperature in  $n$ -type semiconductor

At 0K, both conduction and valence bands are free from any charge carriers and hence, the electrical conductivity is zero.

With increase in temperature, the donor atoms get ionised and hence electron concentration in conduction band increases with temperature until all the donor atoms are ionised. This range is known as **impurity or extrinsic range**.

When the temperature is further increased to room temperature, there are no more donor atoms to be ionised and hence the concentration of electrons in conduction band remains constant over a certain temperature range. This region is known as **exhaustion range**.

As the temperature is increased further, the electrons in valence band are lifted across the forbidden gap to conduction band. Thus, electron concentration increases in conduction band considerably.

With further increase in temperature, more and more electrons from valence band reach conduction band and completely out-number the donor electrons.

The material practically becomes intrinsic and so this range is called **intrinsic range**. All these ranges are shown in fig. 2.13.

The dotted curve indicates hole concentration in an **intrinsic range**.

**Table 2.4**

**Differences between  
intrinsic and extrinsic semiconductors**

| S.No. | Intrinsic semiconductor                  | Extrinsic semiconductor                                                                       |
|-------|------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1.    | It is a pure form of semiconductor.      | An impurity or doping agent is added in the pure semiconductor forms extrinsic semiconductor. |
| 2.    | Number of electrons and holes are equal. | Number of electrons and holes are not equal because of doping.                                |
| 3.    | Conductivity is poor.                    | Conductivity is improved.                                                                     |

**Table 2.5**

**Differences between  
 $n$  - type and  $p$  - type semiconductors**

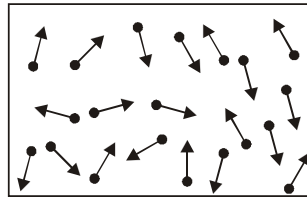
| S.No. | $n$ -type semiconductor                                                                            | $p$ -type semiconductor                                                                          |
|-------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1.    | When pentavalent impurity is doped to intrinsic semiconductor, $n$ - type semiconductor is formed. | When trivalent impurity is doped to intrinsic semiconductor, $p$ - type semiconductor is formed. |

| S.No. | <i>n</i> -type semiconductor                                               | <i>p</i> -type semiconductor                                            |
|-------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 2.    | The impurity is called donor impurity since it donates electron.           | The impurity is called acceptor impurity since it accepts electron.     |
| 3.    | Majority charge carriers are electrons.                                    | Majority charge carriers are holes.                                     |
| 4.    | Minority charge carriers are holes.                                        | Minority charge carriers are electrons.                                 |
| 5.    | The donor energy level is very close to the bottom of the conduction band. | The acceptor energy level is very close to the top of the valence band. |
| 6.    | Fermi energy decreases with increase of temperature.                       | Fermi energy increases with increase of temperature.                    |

## CARRIER TRANSPORT IN SEMICONDUCTOR

### 2.11 RANDOM MOTION & MOBILITY

In absence of an electrical field, the free electrons (electron gas) move in all directions in a random manner. They collide with other free electrons and positive ion core during the motion. This collision is known as **elastic collision** (Fig 2.14).



**Fig. 2.14** *Random motion of free electrons in the absence of electric field (+ ve ion cores are not shown).*

As the motion is random, the resultant velocity in any particular direction is zero.

When an electrical field is applied in a semiconducting material, the free charge carriers such as free electrons and holes attain drift velocity  $v_d$ .

## 2.16.1 Hall Effect

When conductor (metal or semiconductor) carrying a current is placed in a transverse magnetic field, an electric field is produced inside the conductor in a direction normal to both the current and the magnetic field. This phenomenon is known as “**Hall effect**” and the generated voltage is called “**Hall voltage**”.

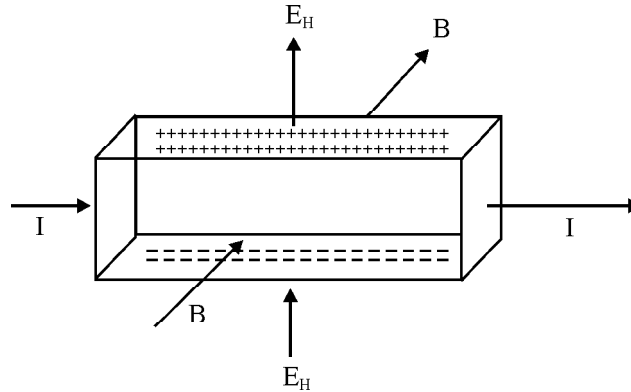


Fig 2.12 Hall effect

## 2.16.2 Hall Effect in n –type Semiconductor

Let us consider an n-type material to which the current is allowed to pass along x-direction from left to right (electrons move from right to left) and the magnetic field is applied in z-directions, as a result Hall voltage is produced in y direction.

Since the direction of current is from left to right the electrons moves from right to left in x-direction as shown in Figure.

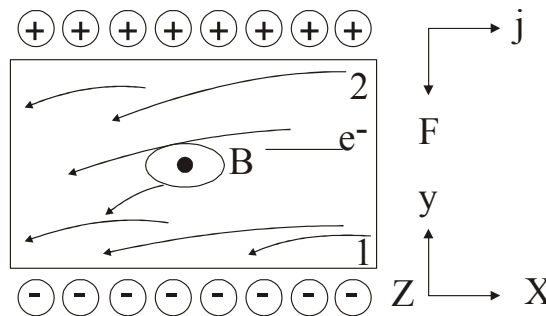


Fig 2.13 Hall effect in N- type semiconductor

Now due to the magnetic field applied the electrons move towards downward direction with the velocity 'v' and cause the negative charge to accumulate at face (1) of the material as shown Figure.

Therefore a potential difference is established between face (2) and face (1) of the specimen which gives rise to field  $E_H$  in the negative y direction.

$$\text{Here, the force due to potential difference} = -eE_H \quad \dots (1)$$

$$\text{Force due to magnetic field} = -Bev \quad \dots (2)$$

At equilibrium eqn. (1) = eqn. (2)

$$-eE_H = -Bev$$

$$E_H = Bv \quad \dots (3)$$

We know the current density  $J_x$  in the x direction is

$$J_x = -n_e ev$$

$$v = \frac{J_x}{N_e e} \quad \dots (4)$$

Substituting eqn. (4) in eqn. (3) we get

$$E_H = \frac{BJ_x}{n_e e} \quad \dots (5)$$

$$E_H = R_H J_x B \quad \dots (6)$$

Where ' $R_H$ ' is known as the Hall coefficient, given by

$$R_H = -(1/n_e e) \quad \dots (7)$$

The negative sign indicates that the field is developed in the negative 'y' direction.

### 2.16.3 Hall Effect in p-type Semiconductor

Let us consider a p-type material for which the current is passed along x-direction from left to right and magnetic field is applied along z-direction as shown in Figure. Since the direction of current is from left to right, the holes will also move in the same direction.

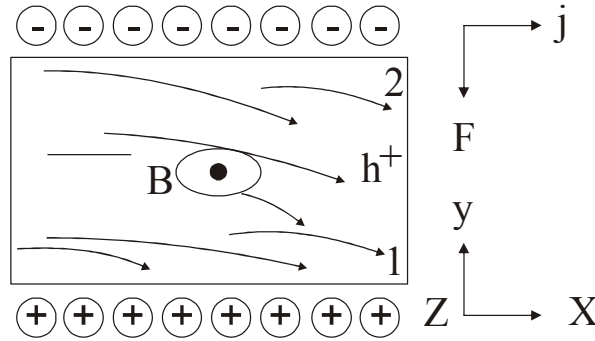


Fig 2.14 Hall effect in P- type semiconductor

Now due to the magnetic field applied, the holes move towards the downward direction with velocity 'v' and accumulate at the face (1) as shown in Figure.

A potential difference is established between face (1) and (2) in the positive y direction.

$$\text{Force due to the potential difference} = eE_H \quad \dots (8)$$

[Since hole is considered to be an electron with same mass but positive charge negative sign is not included].

$$\text{At equilibrium eqn. (7) = eqn. (8)}$$

$$eE_H = Bev$$

$$E_H = Bv$$

$$\text{We known current density } J_x = n_h ev \quad \dots (9)$$

$$v = \frac{J_x}{n_h} e \quad \dots (10)$$

Where  $n_h$  is hole density

Substituting eqn. (10) in (9) we get

$$E_H = \frac{BJ_x}{n_h e}$$

$$E_H = R_H J_x B \quad \dots (11)$$

Where

$$R_H = \frac{1}{n_h e}$$

Equation (11) represents the hall coefficient and the positive sign indicates that the Hall field is developed in the positive y direction.

#### 2.16.4 Hall Coefficient Intermis of Hall Voltage

Half coefficient ( $R_H$ ) is defined as the Hall field developed per unit current density per unit applied magnetic field.

If the thickness of the sample is 't' and the voltage developed is ' $V_H$ ' then Hall voltage

$$V_H = E_H t \quad \dots (12)$$

Substituting eqn. (11) in eqn (12) we have

$$V_H = R_H J_x B t \quad \dots (13)$$

we know Current density  $J_x = \frac{I_x}{A}$

$$\text{Area of the specimen} = \frac{I_x}{bt} \quad \dots (14)$$

where b - is the breath of the sample

t - is the thickness of the sample

Substituting eqn. (14) in eqn (13) we get

$$V_H = \frac{R_H I_x B t}{bt}$$

$$V_H = \frac{R_H I_x B}{b}$$

Hall coefficient

$$R_H = \frac{V_H b}{I_x B}$$

### 2.16.5 Experimental Determination of Hall Effect

A semiconductor slab of thickness 't' and breadth 'b' is taken and current is passed using the battery as shown in Figure.

The slab is placed between the pole of an electromagnet so that current direction coincides with x-axis and magnetic field coincides with z-axis. The hall voltage ( $V_H$ ) is measured by placing two probes at the center of the top and bottom faces of the slab (y-axis).

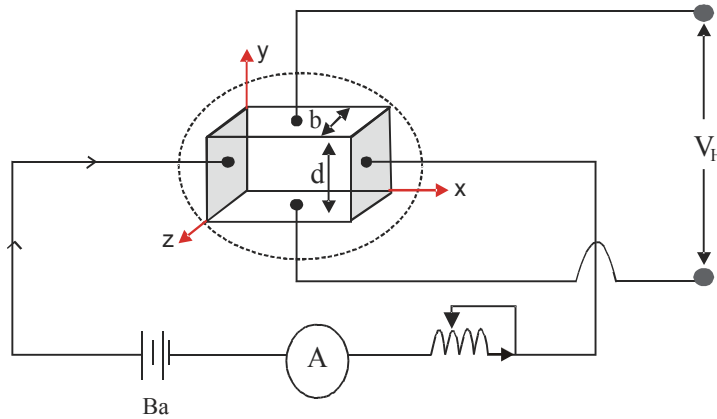


Fig. 2.15 Experimental setup for Hall effect

If  $B$  is magnetic field applied and the  $V_H$  is the Hall voltage produced, then the Hall coefficient can be calculated from the formula

$$R_H = \frac{V_H b}{I_x B} \quad \dots (1)$$

#### Mobility of Charge Carriers

In general the hall co-efficient can be written as

$$R_H = \frac{-1}{ne} \quad \dots (2)$$

The above expression is valid only for conductors where the velocity is taken as the drift velocity. But for semiconductors velocity is taken as average velocity so  $R_H$  for an 'n' type semiconductor is modified as

$$R_H = \frac{-3\pi}{8} \left[ \frac{1}{n_e e} \right]$$
$$R_H = \frac{-1.18}{n_e e} \quad \dots (3)$$

We know the conductivity for n type is  $\sigma = n e \mu_e$

$$\mu_e = \frac{\sigma_e}{n_e e} \quad \dots (4)$$

Eqn. (3) can be rewritten as

$$\frac{1}{n_e e} = \frac{-R_H}{1.18} \quad (5)$$

Substituting eqn. (5) in (4) we get,

$$\mu_e = \frac{-\sigma_e R_H}{1.18} \quad \dots (6)$$

The mobility of electron is in an n-type semiconductor is

$$\mu_e = -\frac{\sigma_e V_H b}{1.18 I \times B} \quad \left[ \because R_H = \frac{V_H b}{I \times B} \right]$$

Similarly for p-type Semiconductor, the mobility of hole is

$$\mu_h = \frac{\sigma_h V_H b}{1.18 (I \times B)} \quad \dots (7)$$

Thus by finding hall voltage, hall coefficient can be calculated and thus the mobility of the charge carriers can also be determined.

### 2.16.6 Application of Hall Effect

1. The sign (N-type (or ) P-type) of charge carriers can be determined.
2. The carrier concentration can be determined  $\left[ n = \frac{1.18}{q R_H} \right]$
3. The mobility of charge carriers is measured directly  $\left[ \mu = \frac{-\sigma R_H}{1.18} \right]$ .
4. Electrical conductivity can be determined  $[\sigma = n q \mu]$ .
5. It can be used to determine whether the given material is metal, insulator, or semiconductor and the type of the semiconductor.
6. It can be used to determine the power flow in an electromagnetic wave.

## SOLVED PROBLEMS

1. Calculate the intrinsic concentration of charge carriers at 300 K given that  $m_e^* = 0.12m_0$ ,  $m_h^* = 0.28m_0$  and the value of band gap = 0.67 eV.

**Solution:**

**Given:**

$$m_e^* = 0.12m_0 = 0.12 \times 9.1 \times 10^{-31} = 1.092 \times 10^{-31} \text{ Kg m}^{-3}$$

$$m_h^* = 0.28m_0 = 0.28 \times 9.1 \times 10^{-31} = 2.548 \times 10^{-31} \text{ Kg m}^{-3}$$

$$T = 300\text{K}.$$

Intrinsic carrier concentration is given by,

$$n_i = 2 \left[ \frac{2\pi kT}{h^2} \right]^{3/2} (m_e^* m_h^*)^{3/4} \exp \left[ \frac{-E_g}{2K_B T} \right]$$

$$2 \left[ \frac{2\pi kT}{h^2} \right]^{3/2} = 2 \left[ \frac{2\pi \times 1.38 \times 10^{-23} \times 300}{6.626 \times 10^{-34}} \right]^{3/2}$$

$$= 2 (1.4421 \times 10^{70})$$

$$= 2.884 \times 10^{70}$$

$$(m_e^* m_h^*)^{3/4} = (1.092 \times 10^{31} \times 2.548 \times 10^{-31})^{3/4}$$

$$= 6.813 \times 10^{-47}$$

$$\exp \left[ \frac{-E_g}{2K_B T} \right] = \exp \left[ - \left( \frac{0.67 \times 1.6 \times 10^{-19}}{2 \times 1.38 \times 10^{-23} \times 300} \right) \right]$$

$$= \exp (-12.9468)$$

$$= 2.3838 \times 10^{-6}$$

$$n_i = [1.442 \times 10^{70}] \times 6.813 \times 10^{-47} \times 2.3836 \times 10^{-6}$$

$$= 2.3407 \times 10^{18}$$

$$\left. \begin{array}{l} \text{Intrinsic carrier} \\ \text{concentration} \end{array} \right\} n_i = 2.3407 \times 10^{18} \text{ m}^{-3}$$

2. The intrinsic carrier density is  $1.5 \times 10^{16} \text{ m}^{-3}$ . If the mobility of electron and hole are 0.13 and  $0.05 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ , calculate the conductivity.

(AU - NOV 2003)

**Solution :**

**Given :**

$$n_i = 1.5 \times 10^{16} \text{ m}^{-3}$$

$$\mu_e = 0.13 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\mu_h = 0.05 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\text{Conductivity } \sigma = n_i e (\mu_e + \mu_h)$$

$$\sigma = 1.5 \times 10^{16} \times 1.6 \times 10^{-19} (0.13 + 0.05)$$

$$\text{Conductivity } \sigma = 4.32 \times 10^{-4} \Omega^{-1} \text{ m}^{-1}$$

3. The Intrinsic carrier density at room temperature in Ge is  $2.37 \times 10^{19} \text{ m}^{-3}$  if the electron and hole mobilities are 0.38 and  $0.18 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$  respectively, calculate the resistivity.

(AU - OCT 2002, DEC 2010)

**Solution:**

**Given:**

$$n_i = 2.37 \times 10^{19} \text{ m}^{-3}$$

$$\mu_e = 0.38 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\mu_h = 0.18 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\text{Conductivity } \sigma = n_i e (\mu_e + \mu_h)$$

$$= 2.37 \times 10^{19} \times 1.6 \times 10^{-19} (0.38 + 0.18)$$

$$= 2.1235 \Omega^{-1} \text{ m}^{-1}$$

$$\text{Resistivity } \rho = \frac{1}{\sigma}$$

$$\rho = \frac{1}{2.1235}$$

$$\text{Resistivity } \rho = 0.4709 \Omega \text{ m}$$

4. The Hall coefficient of certain silicon specimen was found to be  $-7.35 \times 10^{-5} \text{ m}^3 \text{ C}^{-1}$  from 100 to 400 K. Determine the nature of the semiconductor. If the conductivity was found to be  $200 \Omega^{-1} \text{ m}^{-1}$ . Calculate the density and mobility of the charge carrier. (AU - NOV 2002, JUNE 2012)

**Solution:**

**Given:**

$$\text{Conductivity } \sigma = 200 \Omega^{-1} \text{ m}^{-1}$$

$$\text{Hall co-efficient } R_H = -7.35 \times 10^{-5} \text{ m}^3 \text{ C}^{-1} \quad \dots (1)$$

**a) Density of electrons**

$$n = \frac{-1}{R_H e} \text{ (from equation (1))}$$

$$n = \frac{1}{(7.35 \times 10^{-5} \times 1.609 \times 10^{-19})}$$

$$\text{(i.e.)} = 8.455 \times 10^{22} \text{ m}^{-3}$$

We know Conductivity

$$\sigma = n e \mu_e$$

**b) Mobility**

$$\begin{aligned} \mu &= \frac{\sigma}{n e} = \frac{200}{8.455 \times 10^{22} \times 1.6 \times 10^{-19}} \\ &= 0.0147 \end{aligned}$$

$$\text{Mobility } \mu = 0.0147 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$$

$$\text{Density of electrons (n)} = 8.053 \times 10^{22} \text{ m}^{-3}$$

$$\text{Mobility } (\mu) = 0.0147 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$$

5. In a P-type germanium,  $n_i = 2.1 \times 10^{19} \text{ m}^{-3}$  density of boron  $4.5 \times 10^{23} \text{ atoms/m}^3$ . The electron and hole mobility are  $0.4$  and  $0.2 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$  respectively. What is its conductivity before and after addition of boron atoms.

(AU - APRIL 2003)

**Solution:**

**Given:**

$$\text{Intrinsic carrier concentration } n_i = 2.1 \times 10^{19} \text{ m}^{-3}$$

$$\text{Mobility of electrons } \mu_e = 0.4 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$$

$$\text{Mobility of holes } \mu_h = 0.2 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$$

**a) Conductivity before the addition of boron atoms**

$$\begin{aligned}\sigma &= n_i e (\mu_e + \mu_h) \\ &= 2.1 \times 10^{19} \times 1.6 \times 10^{-19} (0.4 + 0.2) \\ &= 2.016 \Omega^{-1} \text{ m}^{-1}\end{aligned}$$

**b) Conductivity after the addition of boron atoms, Boron is a P-type impurity atom**

$$\begin{aligned}\sigma &= p e \mu_h \\ &= 4.5 \times 10^{23} \times 1.6 \times 10^{-19} \times 0.2 \\ \sigma &= 14400 \Omega^{-1} \text{ m}^{-1}\end{aligned}$$

- 6. An N-type semiconductor has hall coefficient =  $4.16 \times 10^{-4} \text{ m}^3 \text{ C}^{-1}$ . The conductivity is  $108 \Omega^{-1} \text{ m}^{-1}$ . Calculate its charge carrier density ' $n_e$ ' and electron mobility at room temperature. (AU - APRIL 2003)**

**Solution:**

**Given:**

$$\text{Hall Co-efficient } R_H = 4.16 \times 10^{-4} \text{ m}^3 \text{ C}^{-1}$$

$$\text{Conductivity } \sigma = 108 \Omega \text{ m}^{-1}$$

1. For 'n' type the charge carriers density  $n_e = \frac{-1}{R_H e}$  Here the negative sign indicates the field direction alone.

$$n_e = \frac{3\pi}{8} \frac{-1}{R_H e}$$

$$n_e = \left[ \frac{3 \times 3.14}{8} \right] \left[ \frac{1}{1.6 \times 10^{-19} \times 4.6 \times 10^{-4}} \right]$$

$$n_e = 1.7690 \times 10^{22} \text{ m}^{-3}$$

2. Electron mobility  $\mu_e = \frac{\sigma_e}{n_e e}$

$$\begin{aligned}&= \frac{108}{\left( 1.7690 \times 10^{22} \times 1.6 \times 10^{-19} \right)} \\ \mu_e &= 0.0381 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}\end{aligned}$$

7. In an N-type semiconductor, the concentration of electron is  $2 \times 10^{22} \text{ m}^{-3}$ . Its electrical conductivity is  $112 \Omega^{-1} \text{ m}^{-1}$ . Calculate the mobility of electrons.

(AU - NOV 2005)

**Solution:**

**Given:**

Conductivity  $\sigma = 112 \Omega^{-1} \text{ m}^{-1}$

Carrier concentration of electron

$$n_i = 2 \times 10^{22} \text{ m}^{-3}$$

Hall coefficient  $R_H = \frac{1}{ne}$

$$= \frac{1}{2 \times 10^{22} \times 1.6 \times 10^{-19}}$$

$$= 3.125 \times 10^{-4} \text{ m}^{-3} \text{ C}^{-1}$$

Mobility  $\mu = \sigma R_H = 112 \times 3.125 \times 10^{-4}$

$$= 0.035 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$$

8. For an intrinsic Semiconductor with a band gap of 0.7 eV, determine the position of  $E_F$  at  $T = 300 \text{ K}$  if  $m_h^* = 6m_e^*$ . (AU - NOV 2003)

**Solution:**

Bandgap  $E_g = 0.7 \text{ eV} = 0.7 \times 1.6 \times 10^{-19} \text{ V}$

$T = 300 \text{ K}$

Fermi energy for an intrinsic semiconductor

$$E_F = \frac{E_g}{2} + \frac{3KT}{4} \log \left[ \frac{m_h^*}{m_e^*} \right]$$

$$E_F = \left[ \frac{0.7 \times 1.6 \times 10^{-19}}{2} \right] + \left[ \frac{3 \times 1.38 \times 10^{-23} \times 300}{4} \right] \log_e 6$$

$$= 6.1563 \times 10^{-20} \text{ Joules}$$

$$E_F = \frac{6.1563 \times 10^{-20}}{1.6 \times 10^{-19}}$$

Fermi energy level  $E_F = 0.3847 \text{ eV}$

9. A semiconducting crystal with 12 mm long, 5 mm wide and 1 mm thick has a magnetic density of  $0.5 \text{ Wbm}^{-2}$  applied from front to back perpendicular to largest faces. When a current of 20 mA flows length wise through the specimen, the voltage measured across its width is found to be  $37\mu\text{V}$ . What is the Hall coefficient of this semiconductor?(AU- DEC 2001, JUNE 2012)

**Solution:**

**Given:**

$$\text{Hall voltage } V_H = 37 \mu\text{V} = 37 \times 10^{-6} \text{ V}$$

$$\text{Breath of the material } t = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$\text{Current } I_H = 20 \text{ mA} = 20 \times 10^{-3} \text{ A}$$

**Magnetic flow density**

$$B = 0.5 \text{ Wbm}^{-2}$$

$$\begin{aligned} \text{Hall coefficient } R_H &= \frac{V_H t}{I_H B} \\ &= \frac{37 \times 10^{-6} \times 1 \times 10^{-3}}{20 \times 10^{-3} \times 0.5} \end{aligned}$$

$$R_H = 3.7 \times 10^{-6} \text{ C}^{-1} \text{ m}^3$$

$$\therefore \text{Hall coefficient } R_H = 3.7 \times 10^{-6} \text{ C}^{-1} \text{ m}^3$$

10. Find the resistance of an intrinsic Ge rod 1 mm long, 1 mm wide and 1 mm thick at 300 K. the intrinsic carrier density  $2.5 \times 10^{19} \text{ m}^{-3}$  is at 300 K and the mobility of electron and hole are 0.39 and  $0.19 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$ .

(AU - APRIL 2003, JUNE 2012)

**Solution:**

**Given:**

$$\text{Length of Ge rod } l = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$\text{Breath } b = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$\text{Thickness } t = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$\text{Intrinsic carrier concentration } n_i = 2.5 \times 10^{19} \text{ m}^{-3}$$

$$\text{Mobility of electron } \mu_e = 0.39 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$$

$$\text{Mobility of hole } \mu_h = 0.19 \text{ m}^2 \text{ v}^{-1} \text{ s}^{-1}$$

**a) Conductivity**

$$\begin{aligned}\sigma &= n_i e (\mu_e + \mu_h) \\ &= 2.5 \times 10^{19} \times 1.6 \times 10^{-19} (0.39 + 0.19) \\ \sigma &= 2.32 \Omega^{-1} \text{ m}^{-1}\end{aligned}$$

**b) Resistance**

$$\begin{aligned}R &= \frac{\ell}{\sigma A} \\ &= \frac{1 \times 10^{-3}}{2.32 \times (1 \times 10^{-3} \times 1 \times 10^{-3})} \quad \therefore (A = b \times t) \\ R &= 431 \Omega\end{aligned}$$

- 11. Hall coefficient of a specimen of doped silicon found to be  $3.66 \times 10^{-4} \text{ m}^3 \text{ C}^{-1}$ . The resistivity of the specimen is  $8.93 \times 10^{-3} \Omega \text{ m}$ . Find the mobility and density of the charge carriers. (AU - APRIL 2004, DEC 2010)**

**Solution:**

$$\text{Hall coefficient } R_H = 3.66 \times 10^{-4} \text{ m}^3 \text{ C}^{-1}$$

$$\text{Resistivity } \rho = 8.93 \times 10^{-3} \Omega \text{ m}$$

**i) Density of holes**

$$\begin{aligned}n_h &= \frac{1}{R_H e} \\ &= \frac{1}{3.66 \times 10^{-4} \times 1.6 \times 10^{-19}} \\ n_h &= 1.7076 \times 10^{22} \text{ m}^{-3}\end{aligned}$$

**ii) Mobility of holes  $\mu_n = \frac{1}{\rho n e}$**

$$\begin{aligned}&= \frac{1}{8.93 \times 10^{-3} \times 1.7076 \times 10^{22} \times 1.6 \times 10^{-19}} \\ \mu_n &= 0.041 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}\end{aligned}$$

12. The intrinsic carrier density of a semiconductor is  $2.1 \times 10^{19} \text{ m}^{-3}$ . The electron and hole mobilities are  $0.4$  and  $0.2 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$  respectively. Calculate the conductivity.

**Solution:**

**Given data:**

$$\text{Intrinsic carrier concentration } n_i = 2.1 \times 10^{19} \text{ m}^{-3}$$

$$\text{Mobility of electron } \mu_e = 0.4 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\text{Mobility of hole } \mu_h = 0.2 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\begin{aligned} \text{Conductivity } \sigma &= n_i e (\mu_e + \mu_h) \\ &= 2.1 \times 10^{19} \times 1.6 \times 10^{-19} \times (0.4 + 0.2) \end{aligned}$$

$$\text{Conductivity } \sigma = 2.016 \Omega^{-1} \text{ m}^{-1}$$

13. The electron mobility and hole mobility in Si are  $0.135 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$  and  $0.048 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$  respectively at room temperature. If the carrier concentration is  $1.5 \times 10^{16} \text{ m}^{-3}$ . Calculate the resistivity of Si at room temperature.

*(AU - JUNE 2009)*

**Solution:**

**Given data:**

$$\text{Carrier concentration } n_i = 1.5 \times 10^{16} \text{ m}^{-3}$$

$$\text{Mobility of electron } \mu_e = 0.135 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\text{Mobility of hole } \mu_h = 0.048 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

**i) Electrical Conductivity  $\sigma$**

$$\begin{aligned} \sigma &= n_i e (\mu_e + \mu_h) \\ &= 1.5 \times 10^{16} \times 1.6 \times 10^{-19} \times (0.135 + 0.048) \end{aligned}$$

$$\sigma = 0.4392 \times 10^{-3} \Omega^{-1} \text{ m}^{-1}$$

**ii) Resistivity of silicon**

$$\begin{aligned} \rho &= \frac{1}{\sigma} \\ &= \frac{1}{0.4392 \times 10^{-3}} \\ \rho &= 2.2768 \Omega \text{ m} \end{aligned}$$

## SHORT QUESTIONS WITH ANSWER

### 1. What is semiconductor?

(AU - April 2002)

The materials in which the electrical conductivity lies between conductors and insulators are called semiconductors. They have resistivity value between  $10^{-4}$  to  $0.5 \Omega\text{m}$ . The electrical conductivity of semiconductor increases, when we added to impurities and by increasing the temperature and it is contrary to the metals. They have negative temperature Coefficient of resistance and which are formed by covalent bonds.

### 2. Distinguish between elemental semiconductors and Compound semiconductors

(AU - Dec 2003, June 2009, Nov 2012)

| S.No | Elemental semiconductors                                                                                    | Compound semiconductors                                                                              |
|------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1.   | These are made from single element.                                                                         | These are made from compound (mixed) element.                                                        |
| 2.   | These are made from IV group and VI group elements                                                          | These are made from III and V [or] II and VI elements.                                               |
| 3.   | These are called as indirect band gap semiconductor (electron-hole recombination takes place through traps) | These are called as direct band gap semiconductor (electron-hole recombination takes place directly) |
| 4.   | Heat is produced in the recombination recombination                                                         | Photons are emitted during                                                                           |
| 5.   | Life time of charge carriers is more due to indirect recombination                                          | Life time of charge carriers is less due to direct recombination.                                    |
| 6.   | Current amplification is more                                                                               | Current amplification is less.                                                                       |
| 7.   | These are used for making diodes, transistor, etc.                                                          | These are used for making LED, laser diodes, etc.                                                    |
| 8.   | Example : Ge, Si                                                                                            | Example : GaAs, GaP, CdS, MgO                                                                        |

**3. What are intrinsic semiconductors?**

A semiconductor in which holes and electrons are created only by thermal excitation across the energy gap is called an intrinsic semiconductor. A pure crystal of silicon or germanium is an intrinsic semiconductor. In an intrinsic semiconductor the number of holes in the valence band is equal to number of electrons in the conduction band. The Fermi level for an intrinsic semiconductor lies at midway in the forbidden gap.

**4. What is meant by compound semiconductor? Give an example.**

*(AU - Nov 2010, June 2011)*

The compound Semiconductor is a semiconductor compound composed of elements from two or more different groups of the periodic table.

i.e., III-V group, II-VI group and IV-VI group.

**Example:** GaAs, GaP, CdS, MgO.

**5. What are extrinsic semiconductors? (or) What is the effect of impurity states over intrinsic semiconductor?**

It is an impure semiconductor made by doping process thereby reducing the band gap up to 0.01 eV.

In this case of N-type semiconductor, the donor energy level is very close to the unfilled energy band (Conduction band). So it can easily donate an electron to that unfilled state.

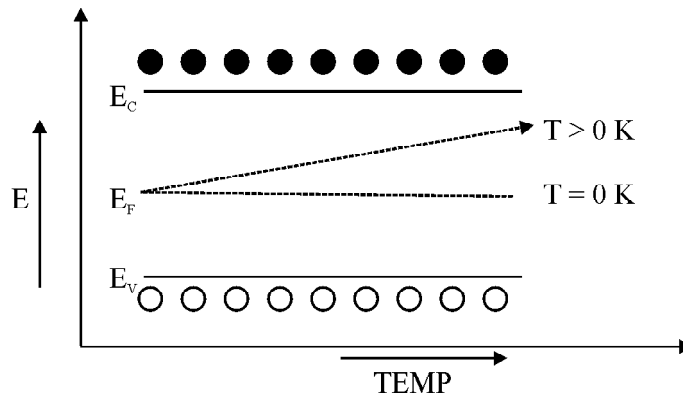
In this case of P-type semiconductor, the acceptor energy level is very close to the filled energy band (Valence band). So it can easily accept the electrons from the filled state.

**6. Differentiate N-type and P-type semiconductor. (AU - Nov 2010, Dec 2009)**

| <i>S. No</i> | <i>N-type</i>                                                           | <i>P-type</i>                                                            |
|--------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1.           | It is donor type                                                        | It is acceptor type                                                      |
| 2.           | Impurity atom is pentavalent                                            | Impurity atom is trivalent                                               |
| 3.           | Donor level lies close to the bottom of the conduction band             | Acceptor level lies close to the top of the valence band.                |
| 4.           | Electrons are the majority carriers and holes are the minority carriers | Holes are the majority carriers and electrons are the minority carriers. |

**7. Sketch the variation of Fermi level in intrinsic semiconductor.**

*(AU - June 201)*



1. At  $T = 0 \text{ K}$ , the Fermi level lies exactly in midway between conduction band and Valence band.
2. At  $T > 0 \text{ K}$ , the Fermi level rises slightly upward since.

**8. What are donor and acceptor impurities?**

A semiconductor in which the impurity atoms are added by doping process is called Extrinsic semiconductor. The addition of impurities increases the carrier concentration and conductivity. There are two types of impurities.

1. Donor impurity which leads to N-type semiconductor.
2. Acceptor impurity which leads to P-type semiconductor.

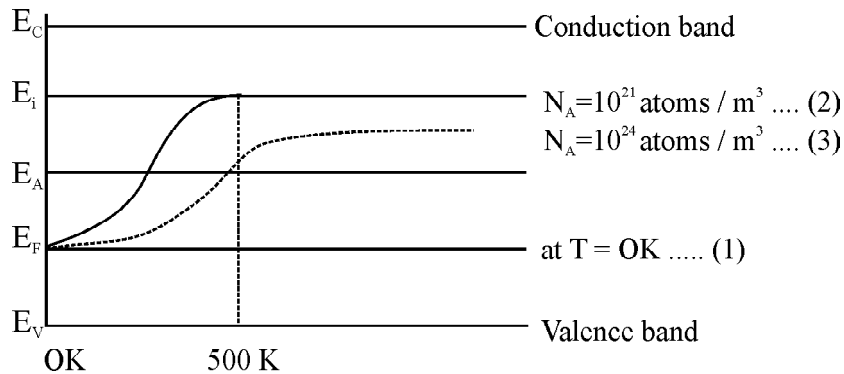
Donor impurity means it donates the electron to the semiconductor materials and Acceptor impurity means it ready to accepts an electron to form the covalent bond in semiconductor materials.

**9. State law of mass action.**

Therefore for intrinsic semiconductor even it impurity is added to increase  $n_e$  there will be decrease in  $n_h$  and hence the product  $n_e n_h$  will remain constant. This is called **Law of mass action**.

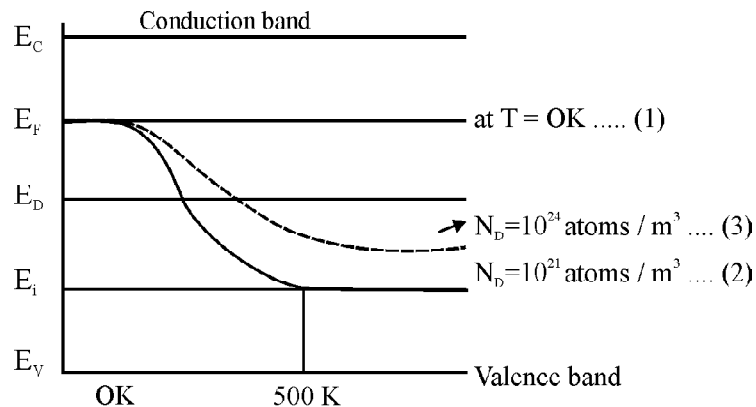
$$\therefore n_e n_h = n_i^2$$

**10. Sketch the variation of Fermi level with temperature for various concentration in P-type semiconductor. (AU - June 2009)**



- 1) When  $T = 0 \text{ K}$ ,  $E_F = \left( \frac{E_V + E_A}{2} \right)$  i.e., at  $T = 0 \text{ K}$ , the Fermi level lies at mid way between the acceptor level and valence level.
- 2) When temperature increases, some of the electrons from valence band will go to acceptor energy level  $[E_A]$ . Therefore the Fermi level shifts upward. At high temperature  $500 \text{ K}$ , the Fermi level reaches intrinsic level  $E_i$ .
- 3) If the impurity atoms are increased from  $10^{21} \text{ atoms/m}^3$  to  $10^{24} \text{ atoms/m}^3$ , the hole concentration increases and hence the Fermi level decrease.

**11. Sketch the variation of Fermi level with temperature for various concentration in N-type semiconductor. (AU - June 2010)**



- 1) When  $T = 0 \text{ K}$ ,  $E_F = \left[ \frac{(E_V + E_D)}{2} \right]$  i.e., at  $T = 0 \text{ K}$ , the Fermi level lies at mid way between the acceptor level and valence level.

- 2) When temperature increases, some of the electrons from valence band will go to acceptor energy level  $[E_A]$ . Therefore the Fermi level shifts upward. At high temperature 500 K, the Fermi level reaches intrinsic level  $E_i$ .
- 3) If the impurity atoms are increased from  $10^{21} \text{ atoms / m}^3$  to  $10^{24} \text{ atoms / m}^3$ , the hole concentration increases and hence the Fermi level decrease.

## 12. What is pair production or generation?

When an energy is supplied to the semiconductor, the covalent bonds are broken and the electrons are raised from valence band to conduction band and a vacant site is created in the valence band. It is called pair production or generation of electron-hole pair.

## 13. What is Recombination?

Recombination is the conversion of a free electron into a bound electron by occupying a vacant site. An electron jumps from the conduction band to the valence band vacant site is called as band to band recombination of electron-hole pair.

During band to band combination the excess energy is released in the form of light photon. The recombination also occurs via recombination centre and traps.

## 14. Define Hall effect? *(AU - May 2004, 2007, Dec 2009, June 2009)*

If a semiconductor or a conductor carrying current (I) is placed in a magnetic field (B), an electric field is produced in the direction normal to both the current and magnetic field. The generated voltage is called the Hall voltage ( $V_H$ ) and the field is called Hall field ( $E_H$ ).

## 15. What are the applications of hall effect? *(AU - June 2010, 2012)*

1. The sign (N-type (or ) P-type) of charge carriers can be determined.
2. The carrier concentration can be determined  $\left[ n = \frac{1.18}{qR_H} \right]$
3. The mobility of charge carriers in measured directly  $\left[ \mu = \frac{-\sigma R_H}{1.18} \right]$ .
4. Electrical conductivity can be determined  $[\sigma = n q \mu]$ .
5. It can be used to determine whether the given material is metal, insulator, or semiconductor and the type of the semiconductor.
6. It can be used to determine the power flow in an electromagnetic wave.

**16. Give on extrinsic semiconductor. How will you find whether it is n-type (or) P type. (AU - June 2010, 2012)**

If pentavalent (Phosphorous, Arsenic, Antimony) impurities are doped with pure semiconducting material the free electrons are produced, this is called n-type semiconductor. In n-type semiconductor majority charge carriers is electron.

If trivalent [Aluminium, Gallium, Indium] impurities are doped with pure semiconducting material the holes are produced, this is called p - type semiconductor. In p - type semiconductor the majority charge carrier is holes.

**17. Define diffusion current**

In addition to drift current, there is also another current called diffusion current. Diffusion current does not takes place in metals.

Diffusion current is defined as the motion of charge carriers from the region of higher concentration to the region of lower concentration.

**18. State the properties of Semiconductors. (AU - May 2003, June 2010)**

1. They are formed by covalent bonds.
2. They have small energy gap.
3. They have an empty conduction band and almost filled valence band 0K.
4. They have negative temperature coefficient of resistance.
5. They resistivity of semiconductors lies between a semiconductor and Insulator. ( $10^{-4}$  to 0.5 Ohms m).
6. At 0 K it behave as insulator.

**19. What are the differences between conductor and a semiconductor?**

| <i>S.No</i> | <i>Conductor</i>                                         | <i>Semiconductor</i>                                                                                |
|-------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1.          | The conductor is a material which has low resistivity    | The semiconductor is a material which has resistivity lying between the conductor and an insulator. |
| 2.          | They will not behave as an insulator at any temperature. | The pure form of semiconductor can behave as an insulator at 0K.                                    |
| 3.          | They have positive coefficient of resistance             | They have negative temperature coefficient of resistance.                                           |

**20. Why do we prefer silicon for transistor and Ga As for Laser diodes?**

*(AU - June 2009)*

Silicon is an indirect band gap semiconductor for which the life time of the charge carriers is more and the current amplification is also very high. Hence it is preferable for use in transistor.

Ga As is a direct band gap semiconductor, in which electrons and holes recombine directly to produce photons and hence used in laser diodes.

**21. What happens when the temperature increase in the case of semiconductor and conductor?**

*(AU - June 2010)*

With increase of temperature the conductivity of semiconductor increases and hence resistivity decreases because more and more charge carriers are created by the temperature.

**22. Write the expression for electrical conductivity of an intrinsic semiconductor.**

*(AU - Dec 2012)*

Electrical conductivity  $\sigma = n_i e (\mu_e + \mu_h)$

Where  $n_i$  - is the intrinsic carrier concentration

$\mu_e$  - is the mobility of electrons

$\mu_h$  - is the mobility of holes

**23. Give the carrier concentration of an intrinsic semiconductor. (AU - Jun 2010)**

The carrier concentration in an intrinsic semiconductor is

$$n_i = 2 \left( \frac{2\pi k_B T}{h^2} \right)^{\frac{3}{2}} (m_e^* m_h^*)^{3/4} e^{-E_g / 2K_B T}$$

**24. Write the expression for energy gap ( $E_g$ ) of an intrinsic semiconductor.**

*(AU - May 2011)*

Energy gap of a semiconductor  $E_g = 2K_B \left( \frac{dy}{dx} \right)$

$\left( \frac{dy}{dx} \right)$  - is the slope of the curve

## **PART - B QUESTIONS**

1. Assuming Fermi-Dirac statistics, derive expressions for the density of electrons and holes in an intrinsic semiconductor. Hence, obtain expression for the electrical conductivity of the intrinsic semiconductor. *(AU - April 2002)*
2. Derive an expression for,
  - i) Density of electrons in the conduction band in an intrinsic semiconductor.
  - ii) Density of holes in the valance band in an intrinsic semiconductor.
3. Describe the method of determining the band gap of a semiconductor. How does the electrical conductivity vary with temperature for an intrinsic semiconductor. *(AU - Nov 2003)*
4. Obtain an expression for, density of electrons in the conduction band of an N-type semiconductor and density of holes in the conduction band of an P-type semiconductor by assuming Fermi-Dirac distribution function. *(AU - May 2004)*
5. Discuss the variation of carrier concentration with temperature in n-type semiconductor.
6. Derive an expression in intrinsic semiconductor.
7.
  - i) Derive an expression for the carrier concentration in intrinsic semiconductor.
  - ii) Explain the variation of electrical conductivity with respect to temperature in the case of an intrinsic semiconductor. *(AU - June 2006)*
8. Describe a method of determining the band gap energy of a semiconductor.
9. Derive an expression for carrier concentration with temperature N-type semiconductor, and describe an experimental set-up for the determination of Hall Co-efficient and Hall voltage. *(AU - May 2004)*
10. Explain the variation of Fermi level with temperature *(AU - June 2006)*
11. Explain the variation of Fermi level with temperature and impurity concentration in N-type semiconductor.
12. What is Hall effect? Describe an experiment for the measurement of Hall coefficient, and write its applications. *(AU - May 2007)*
13. Derive an expression for density of holes in the valance band and also explain how does the Fermi level vary with concentration of impurities in P-type semiconductor. *(AU - June 2006)*

## ASSIGNMENT PROBLEMS

1. Find the resistance of an intrinsic germanium rod 1 cm long, 1mm wide and 1mm thick at 300 K. the intrinsic carrier density is  $2.5 \times 10^{19} / \text{m}^3$  at 300 K and the mobility of electron and hole are 0.39 and  $0.19 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$ . **(Ans:  $4.31 \times 10^3 \Omega$ )**
2. Calculate the position of Fermi level  $E_F$  and the conductivity at 300 K for germanium crystal containing  $5 \times 10^{22}$  arsenic atoms /  $\text{m}^3$ . Also calculate the conductivity if the mobility of the electron is  $0.39 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$ .  
**(Ans :  $E_F$  is 0.16 eV below  $E_c$ ,  $\sigma = 3210 \Omega^{-1} \text{ m}^{-1}$ )**
3. In a Hall experiment a current of 25 A is passed through a long foil of silver which is 0.1mm thick and 3cm wide. If the magnetic field of flux density  $0.14 \text{ Wb/m}^2$  is applied perpendicular to the foil, calculate the Hall voltage development and estimate the mobility of electrons in silver. The conductivity the Hall coefficient is  $(-8.4 \times 10^{-11}) \text{ m}^3 / \text{coulomb}$ . **(Ans :  $29.4 \mu\text{V}$  and  $57.7 \times 10^{-4} \text{ m}^2 \text{ V}^{-1}$ )**
4. The intrinsic carrier density at room temperature in Ge is  $2.37 \times 10^{19} \text{ m}^3$ . If the electron and hole motilities are 0.38 and  $0.18 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$  respectively, calculate the resistivity.  
**(Ans :  $0.471 \Omega\text{m}$ )**
5. For silicon semiconductor with band gap 1.12 eV, determine the position of the Fermi level at 300 K, if  $m_e^* = 0.12m_0$  and  $m_h^* = 0.28m_0$  **(Ans : 0.576 eV)**
6. For an intrinsic semiconductor with gap width  $E_g = 0.7 \text{ eV}$ , calculate the concentration of intrinsic charge carriers at 300 K assuming that  $m_e^* = m_h^* = m_0$ .  
**(Ans :  $33.49 \times 10^{18} / \text{m}^3$ )**
7. A silicon plate of thickness 1mm, breadth 10mm, and length 100mm is placed magnetic field of  $0.5 \text{ wb/m}^2$  acting perpendicular to its thickness. If  $10^{-2}$  current flows along its length, calculate the Hall voltage developed if the Hall coefficient is  $3.66 \times 10^{-4} \text{ m}^3 / \text{coulomb}$ . **(Ans :  $3.7 \times 10^6 \text{ C}^{-1} \text{ m}^3$ )**
8. A N-type semiconductor has Hall coefficient  $= 4.16 \times 10^{-4} \text{ C}^{-1} \text{ m}^3$ . The conductivity is  $108 \text{ ohm}^{-1} \text{ m}^{-1}$ . Calculate its charge carrier density and electron mobility at room temperature.  
**(Ans :  $0.038 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$ )**